Simplicial Semantics for Intuitionistic Logic: A No-Go Theorem and Geometric Intermediate Logics

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Abstract

We develop the semantic foundations of SC-Logic, a propositional logic in which formulas denote subcomplexes of a finite simplicial complex. The lattice of subcomplexes carries a canonical Heyting-algebra structure, and we give explicit descriptions of its implication and Heyting negation. Monotone filtrations induce finite Kripke frames whose worlds are filtration stages and whose valuations are persistent; for discrete filtrations, the resulting simplex-level forcing semantics agrees with the corresponding algebraic semantics. We establish soundness and strong completeness of a Hilbert-style proof system for the class of filtration models considered here, together with the finite model property and decidability of the resulting modality-free logic.

Our main result is a no-go theorem for the modality-free fragment of the language. It states that, for every realising class of finite monotone filtrations, the formulas valid throughout the class are exactly the theorems of intuitionistic propositional logic. Consequently, no monotone constructible condition imposed solely on the stage complexes, including connectedness, dimension bounds, contractibility, or analogous geometric restrictions, can yield a proper superintuitionistic logic in this setting. This conclusion does not rule out restrictions on the filtration poset itself. In particular, bounded-depth restrictions on the poset give rise to proper intermediate logics; for the corresponding classes, the resulting logic is the bounded-depth intermediate logic BD_n. Thus, geometric restrictions on the stages and order-theoretic restrictions on the filtration must be distinguished when studying extensions of intuitionistic logic. We also investigate an enrichment by a spatial modality defined through star closure. The resulting operator is normal and reflexive but not transitive, with the failure of transitivity arising from boundary effects. The modal analysis is exploratory: no general complete-

ness theorem for the enriched language is claimed. Finally, we distinguish set-theoretic complement from Heyting negation, introduce a filtration-stage-indexed graded negation, and study the clarity operator given by the double-negation nucleus. Its fixed points are the regular subcomplexes, which form the Booleanization of the underlying Heyting algebra. These results identify the modality-free propositional core of topology-aware reasoning in the present simplicial semantics with intuitionistic logic, while clarifying the separate roles of geometric, order-theoretic, and modal structure.

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1. Introduction

Since Tarski's topological interpretation of the intuitionistic propositional calculus [30, 23], one strand of logical semantics has held that the carrier of truth need not be an abstract set of valuations: it can be a space, with the connectives read off from its topology. The open sets of any topological space form a Heyting algebra, and intuitionistic logic is exactly the logic of that reading. The idea has since grown into a systematic *spatial logic* [1, 2] and, in the finite setting, into the correspondence between finite spaces, posets and their order complexes [22, 3].

A parallel development took place outside logic. In topological data analysis the basic object is not a single space but a *filtration*: a monotone family of simplicial complexes indexed by a parameter, read categorically as a functor on a poset [5, 7]. Filtered complexes are by now the standard data structure for geometric information, and sheaf-theoretic refinements carry the same idea to distributed and sensor data [8]. Independently again, simplicial complexes have entered logic proper as models of multi-agent epistemic reasoning, with vertices as local states, simplices as consistent global configurations, and complete axiomatisations for the resulting semantics [16, 10].

Between these traditions lies a gap that is easy to state. Spatial logic assigns truth values to *open sets*; the epistemic-simplicial reading assigns

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worlds to simplices and uses the complex as an indexing device for agents. Neither interprets a formula as a *subcomplex*, a face-closed subobject, and lets the face relation itself do the logical work. Two consequences follow immediately. Subcomplexes are closed under union and intersection but not under complementation, so negation cannot be set-theoretic; and a filtration hands us an accessibility structure along which truth may be required to persist. What logic do these two features determine, and how much of it is genuinely geometric? Those are the questions of this paper. One might expect geometry to impose further logical constraints, because face-closure and persistent growth restrict the admissible truth sets.

Paper I of this series [25] introduced *SC-Logic*, in which every formula denotes a subcomplex of a finite complex K, and fixed the object language together with its simplex-level valuations. It left the semantics and the proof theory implicit. We supply them here, and the answer comes in two halves.

The logic is exactly IPC, and geometry cannot change that.. SC-Logic is sound and strongly complete for the class of filtration models (Theorems 4.5 and 4.8). Crucially, no geometric hypothesis on the stages, when considered within a realising class in the sense of Definition 5.1, yields a proper superintuitionistic logic for the base language $\mathcal{L}_0$ (Theorem 4.20). We insist that this second half is a theorem and not a caveat. It pins down IPC as the exact propositional invariant of the propositional, modality-free fragment of spatial reasoning over finite simplicial complexes, telling anyone designing such a system which features can affect validity and which cannot. The precedent is familiar: completeness of $\mathsf{S4}$ for a single space such as $\mathbb{R}$ [23] counts as a strength of that semantics, not a defect.

The algebra of subcomplexes, explicitly.. Section 2 shows that $\mathrm{Sub}(K)$ is a Heyting algebra whose implication is the largest subcomplex C with $A \cap C \subseteq B$ (Proposition 2.11), and identifies the algebra up to isomorphism as the down-sets of the face poset, $\mathrm{Sub}(K) \cong \mathrm{Down}(\mathcal{F}(K))$ (Theorem 2.15). This representation explains the inertness phenomenon by placing the semantics inside the Alexandrov reading of finite posets. It also constrains which algebras are realisable at all: no complex, for instance, has a three-element subcomplex lattice. Heyting negation is computed by *coface removal* and is in general strictly smaller than the set-theoretic complement (Proposition 2.12).

Filtrations as Kripke frames.. To a monotone family $(K_t)_{t\in T}$, Section 3 attaches a simplex-level forcing relation $\Vdash_\sigma$ with persistent valuations (Definitions 3.4 and 3.3). An agreement theorem relates this forcing to the algebraic semantics; the two coincide on one-stage filtrations and diverge in general, so the readings must not be traded for one another (Proposition 3.8). Completeness is proved by a *point filtration* (Definition 4.10), which transports an arbitrary intuitionistic countermodel into the geometric setting. The finite model property follows with the explicit stage bound $2^{|\mathrm{Sub}(\varphi)|}$ (Proposition 4.15); the finiteness of the relevant free algebras is Diego's theorem [9, 6].

Where the geometry does show up.. Section 6 turns from validity to content. We introduce stage-indexed *graded negations* $\neg^{(t)}$ (Definition 6.3), which measure how persistently a subcomplex avoids a region, and the *clarity operator* $\mathrm{Cl}\colon A \mapsto \neg_H\neg_H A$ (Definition 6.6). Clarity is the semantic counterpart of the double-negation nucleus isolated by Glivenko [15]; it adds no validities to $\mathcal{L}_0$, but its fixed points form the Booleanisation of $\mathrm{Sub}(K)$, which we characterise as the *regular* subcomplexes. The graded negations are stage-local and not internalizable in $\mathcal{L}_0$ (Proposition 6.11), living instead on the algebras $\mathrm{Sub}(K_t)$. This lack of internal expressivity motivates the introduction of the spatial modality $\Box$ in Section 9.

Antecedents and Modal Perspective.. The logical side descends from Brouwer and its formalisation by Heyting [18], consolidated by Rasiowa and Sikorski [27] and surveyed by Troelstra and van Dalen [31]. We add a *combinatorial realisation* where the nucleus vocabulary of locale theory [19] and bi-Heyting refinements [21, 28] receive a geometric grounding. Regarding modalities, adding $\Box$ to intuitionistic logic has a long history [24, 4]. Our modality in Section 9 is determined by the face relation of the stage complex. While normality and T hold, $\mathsf{4}$ fails because the interior operator intervenes between successive applications of the coface passage (Propositions 9.4 and 9.10).

Outline.. Section 2 develops the Heyting structure of $\mathrm{Sub}(K)$. Section 3 introduces filtration frames and forcing. Section 4 gives the proof system with soundness and strong completeness, and Section 5 proves the no-go results for $\mathcal{L}_0$. Section 6 treats graded negation and clarity; Section 7 runs a worked case; Section 8 situates SC-Logic among related systems; and Section 9 develops the spatial modality $\Box$. Section 10 states the open problems, particularly

regarding the characterisation of logics arising from non-degenerate filtration models.

2. The Heyting Algebra of Subcomplexes

The aim of this section is to show that for a fixed finite simplicial complex K the collection $\mathrm{Sub}(K)$ of its subcomplexes is a complete Heyting algebra, and to identify that algebra up to isomorphism. We first fix the subcomplex viewpoint and the interior/closure operators, then develop the lattice operations and the Heyting implication on $\mathrm{Sub}(K)$, then prove the face-poset representation theorem together with the realisability constraint it imposes, and finally introduce valuations as the semantic bridge from syntax to complexes. The running example throughout is the boundary complex $\partial\Delta^2$.

Notational conventions.. Lowercase Latin letters $p, q, r, \ldots$ always denote *propositional variables*; uppercase $A, B, C, \ldots$ always denote *subcomplexes* of the ambient complex K. The two are linked only through a valuation (Definition 2.21); no formula is ever identified with a subcomplex. We reserve $\bot, \top$ for the syntactic constants and write $0, 1$ for the bounds of $\mathrm{Sub}(K)$.

2.1. Preliminaries

Language scope.. We fix two languages. The *base language* $\mathcal{L}_0$ is the set of $\Box$-free formulas generated by

$$\varphi ::= p \mid \bot \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \to \varphi,$$

where p ranges over a countable set Prop of atomic propositions; we abbreviate $\neg\varphi := \varphi \to \bot$ and $\top := \bot \to \bot$. The *extended language* $\mathcal{L}_\Box$ is $\mathcal{L}_0$ augmented by the formation clause

$$\varphi \in \mathcal{L}_\Box \implies \Box\varphi \in \mathcal{L}_\Box.$$

Thus $\mathcal{L}_0 \subsetneq \mathcal{L}_\Box$, and $\mathcal{L}_\Box$ is closed under all connectives and under arbitrary nesting of $\Box$; in particular $\Box\Box\varphi \in \mathcal{L}_\Box$, which is needed even to state the axiom $\mathsf{4}$ in Section 9. Unless stated otherwise, every result in Sections 2–6 concerns $\mathcal{L}_0$-formulas only; the spatial modality $\Box$ is treated separately in Section 9.

The next definition is the standard starting point: every geometric argument in this paper is combinatorial and must preserve faces.

Definition 2.1 (Abstract simplicial complex, cf. [17])**.** Let V be a finite set of *vertices*. A (finite, abstract) *simplicial complex* K over V is a collection of non-empty subsets of V, called *simplices*, such that

1. if $\sigma \in K$ and $\emptyset \neq \tau \subseteq \sigma$ then $\tau \in K$ (closure under non-empty faces), and
2. $\bigcup_{\sigma \in K} \sigma = V$.

The *dimension* of $\sigma \in K$ is $|\sigma| - 1$, and $\dim K = \max_{\sigma \in K} \dim \sigma$.

Definition 2.2 (Subcomplexes)**.** Let K be a finite simplicial complex. A set $A \subseteq K$ is *face-closed* if $\sigma \in A$ and $\emptyset \neq \tau \subseteq \sigma$ imply $\tau \in A$. Put

$$\mathrm{Sub}(K) \;:=\; \{\, A \subseteq K \;\mid\; A \text{ is face-closed} \,\}.$$

Its elements are the *subcomplexes* of K. Both $\emptyset$ and K belong to $\mathrm{Sub}(K)$, and a subcomplex need not contain every vertex of K: condition (2) of Definition 2.1 normalises the *ambient* complex only and is not imposed on its subcomplexes. Every non-empty $A \in \mathrm{Sub}(K)$ is itself a simplicial complex over its own vertex set $\bigcup_{\sigma \in A} \sigma$.

Remark 2.3. Admitting $\emptyset$ as a subcomplex is exactly what makes $\mathrm{Sub}(K)$ a bounded lattice, with $\emptyset$ interpreting $\bot$. Every occurrence of "subcomplex" below is in the sense of Definition 2.2.

Definition 2.4 (Interior and closure)**.** Let L be a simplicial complex and $S \subseteq L$ an arbitrary set of simplices. Put

$$\mathrm{int}_L(S) = \{\sigma \in S \mid \emptyset \neq \tau \subseteq \sigma \Rightarrow \tau \in S\}, \qquad \mathrm{cl}_L(S) = \{\tau \in L \mid \exists\, \sigma \in S,\ \tau \subseteq \sigma\}.$$

We drop the subscript when the ambient complex is the fixed K. A subset $U \subseteq L$ is *face-open* in L if $U = \mathrm{int}_L(U)$, equivalently if $U \in \mathrm{Sub}(L)$. That $\mathrm{int}_L(S)$ is the *largest* subcomplex contained in S, and $\mathrm{cl}_L(S)$ the *smallest* subcomplex containing S, is proved in Lemma 2.6; it is not part of the definition.

Example 2.5. Let $K = \partial\Delta^2$ on vertices $\{a, b, c\}$:

$$K = \big\{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}\big\}.$$

The three vertices and three edges form a 1-dimensional cycle, with no 2-simplex. This fixed K is used for examples throughout the remainder of the paper.

2.2. The Heyting algebra structure

Lemma 2.6 (Closure and extremality)**.** *Let K be a finite simplicial complex and $S \subseteq K$ arbitrary.*

1. $\mathrm{Sub}(K)$ *is closed under arbitrary intersections and arbitrary unions; in particular* $A \cap B$, $A \cup B \in \mathrm{Sub}(K)$ *for all* $A, B \in \mathrm{Sub}(K)$.

2. $\mathrm{int}(S)$ *is the largest element of* $\mathrm{Sub}(K)$ *contained in* S*, and* $\mathrm{cl}(S)$ *the smallest element of* $\mathrm{Sub}(K)$ *containing* S.

Proof. (1) Let $\{A_i\}_{i\in I} \subseteq \mathrm{Sub}(K)$ and $\emptyset \neq \tau \subseteq \sigma$. If $\sigma \in \bigcap_i A_i$ then $\tau \in A_i$ for every i; if $\sigma \in \bigcup_i A_i$, choose i with $\sigma \in A_i$ and then $\tau \in A_i$. Both cases hold vacuously for $I = \emptyset$ under the conventions $\bigcap_\emptyset = K$, $\bigcup_\emptyset = \emptyset$.

(2) *Interior.* Face-closure: if $\sigma \in \mathrm{int}(S)$ and $\emptyset \neq \tau \subseteq \sigma$, then every non-empty $\rho \subseteq \tau$ is a face of σ and hence lies in S, so $\tau \in \mathrm{int}(S)$. Containment $\mathrm{int}(S) \subseteq S$ is immediate. Maximality: let $A \in \mathrm{Sub}(K)$ with $A \subseteq S$ and $\sigma \in A$; every non-empty $\tau \subseteq \sigma$ lies in A by face-closure of A, hence in S, so $\sigma \in \mathrm{int}(S)$ and $A \subseteq \mathrm{int}(S)$.

Closure. Face-closure: if $\tau \subseteq \sigma \in S$ and $\emptyset \neq \rho \subseteq \tau$ then $\rho \subseteq \sigma$, so $\rho \in \mathrm{cl}(S)$. Containment $S \subseteq \mathrm{cl}(S)$ is immediate. Minimality: if $S \subseteq A$ with A face-closed and $\tau \subseteq \sigma \in S \subseteq A$, then $\tau \in A$, so $\mathrm{cl}(S) \subseteq A$. □

Definition 2.7 (Order and constants)**.** Order $\mathrm{Sub}(K)$ by inclusion and set

$$A \wedge B := A \cap B, \qquad A \vee B := A \cup B, \qquad 0 := \emptyset, \qquad 1 := K,$$

which is well defined by Lemma 2.6(1).

Meet and join are the facewise geometric operations, and the constants record truth and falsity across every simplex. Implication cannot be read off pointwise in the same way, and it is there that the geometry begins to bite.

Definition 2.8 (Heyting implication)**.** For $A, B \in \mathrm{Sub}(K)$ define

$$A \to_H B \;:=\; \mathrm{int}\big((K \setminus A) \cup B\big), \qquad \neg_H A \;:=\; A \to_H 0 \;=\; \mathrm{int}(K \setminus A).$$

Remark 2.9. The set difference $K \setminus A$, and more generally $(K \setminus A) \cup B$, need not be face-closed and so is in general *not* a subcomplex. Applying int repairs this by passing to the largest subcomplex contained in the given set of simplices (Lemma 2.6(2)). This single repair is the only reason int occurs in Definition 2.8, and the gap it opens between $\neg_H A$ and $K \setminus A$ is the source of everything that follows.

Lemma 2.10 (Residuation)**.** *For all $A, B, C \in \mathrm{Sub}(K)$,*

$$C \cap A \subseteq B \iff C \subseteq A \to_H B.$$

Consequently $A \to_H B = \bigcup\{\, C \in \mathrm{Sub}(K) \mid C \cap A \subseteq B \,\}$ is the largest subcomplex C with $C \cap A \subseteq B$.

Proof. ($\Rightarrow$) Assume $C \cap A \subseteq B$ and let $\sigma \in C$. If $\sigma \notin A$ then $\sigma \in K \setminus A$; otherwise $\sigma \in C \cap A \subseteq B$. Hence $C \subseteq (K \setminus A) \cup B$, and since C is a subcomplex, Lemma 2.6(2) gives $C \subseteq \mathrm{int}((K \setminus A) \cup B) = A \to_H B$.

($\Leftarrow$) If $C \subseteq A \to_H B \subseteq (K \setminus A) \cup B$, then any $\sigma \in C \cap A$ lies in $(K \setminus A) \cup B$ and in A, hence in B.

For the displayed formula, take $C = A \to_H B$ in ($\Leftarrow$) to see that $A \to_H B$ itself satisfies the condition, and ($\Rightarrow$) shows it contains every C that does. □

Proposition 2.11. *The structure $(\mathrm{Sub}(K), \subseteq, \cap, \cup, \to_H, \emptyset, K)$ is a Heyting algebra. It is complete, and satisfies the infinite distributive law $A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i)$; that is, it is a frame.*

Proof. By Lemma 2.6(1), $(\mathrm{Sub}(K), \subseteq)$ is a lattice with binary meet $\cap$ and join $\cup$, bounded by $\emptyset$ and K and closed under arbitrary meets and joins, hence complete. It is closed under $\to_H$ because int takes values in $\mathrm{Sub}(K)$ by Lemma 2.6(2); note that $(K \setminus A) \cup B$ is in general *not* face-closed, which is precisely why the interior is applied (Remark 2.9). Lemma 2.10 supplies the adjunction $C \cap A \subseteq B \iff C \subseteq A \to_H B$, so $\to_H$ is the relative pseudo-complement and $\mathrm{Sub}(K)$ is a Heyting algebra. The infinite distributive law holds because meets and joins are computed as set-theoretic intersections and unions, where it is a tautology. □

Proposition 2.12 (Coface removal computes Heyting negation)**.** *For every $A \in \mathrm{Sub}(K)$,*

$$\neg_H A \;=\; \mathrm{int}(K \setminus A) \;=\; K \setminus \{\sigma \in K \mid \exists\, \tau \in A,\ \tau \subseteq \sigma\}.$$

The right-hand side is already face-closed, so no further closure operator is required: Heyting negation deletes A together with all of its cofaces.

Proof. Write $U = \{\sigma \in K \mid \exists\, \tau \in A,\ \tau \subseteq \sigma\}$ for the set of cofaces of simplices of A. Since $A \subseteq U$, we get $K \setminus U \subseteq K \setminus A$. The set $K \setminus U$ is face-closed: if $\sigma \in K \setminus U$ and $\emptyset \neq \tau \subseteq \sigma$, then $\tau \in U$ would give some $\rho \in A$ with $\rho \subseteq \tau \subseteq \sigma$, forcing $\sigma \in U$, a contradiction. So $K \setminus U$ is a subcomplex contained in $K \setminus A$,

whence $K \setminus U \subseteq \mathrm{int}(K \setminus A)$ by Lemma 2.6(2). Conversely let $\sigma \in \mathrm{int}(K \setminus A)$. If $\sigma \in U$, choose $\tau \in A$ with $\tau \subseteq \sigma$; face-closure of $\mathrm{int}(K \setminus A)$ forces $\tau \in K \setminus A$, contradicting $\tau \in A$. Hence $\sigma \in K \setminus U$. The two inclusions give the claim. □

Corollary 2.13 (Heyting negation)**.** *For every $A \in \mathrm{Sub}(K)$,*

$$\neg_H A \;=\; A \to_H 0 \;=\; \bigcup \{\, C \in \mathrm{Sub}(K) \mid C \cap A = \emptyset \,\} \;=\; \mathrm{int}(K \setminus A),$$

so $\neg_H A$ is the largest subcomplex of K disjoint from A. Moreover $\neg_H A = K \setminus A$ if and only if $K \setminus A$ is face-closed, equivalently if and only if A is closed under cofaces in K.

Proof. The displayed identities are Definition 2.8 together with Lemma 2.10 at $B = \emptyset$. If $K \setminus A$ is face-closed then $\mathrm{int}(K \setminus A) = K \setminus A$; conversely int always returns a face-closed set. Face-closure of $K \setminus A$ says that $\sigma \in K \setminus A$ and $\tau \subseteq \sigma$ imply $\tau \in K \setminus A$, whose contrapositive is that $\tau \in A$ and $\tau \subseteq \sigma$ imply $\sigma \in A$, i.e. coface-closure of A. □

Example 2.14. In $K = \partial\Delta^2$ from Example 2.5 take $A = \big\{\{a\}, \{b\}, \{a, b\}\big\}$. Then

$$K \setminus A = \big\{\{c\}, \{a, c\}, \{b, c\}\big\}, \qquad \neg_H A = \mathrm{int}(K \setminus A) = \big\{\{c\}\big\}.$$

Only the vertex $\{c\}$ survives: each of $\{a, c\}$ and $\{b, c\}$ has a face inside A (namely $\{a\}$, resp. $\{b\}$), so both are cofaces of A and are deleted by Proposition 2.12. Here $A \vee \neg_H A = \big\{\{a\}, \{b\}, \{c\}, \{a, b\}\big\} \subsetneq K$, witnessing the failure of excluded middle.

2.3. Representation and realisability

Theorem 2.15 (Face-poset representation of $\mathrm{Sub}(K)$)**.** *Let K be a finite simplicial complex and $\mathcal{F}(K)$ its face poset, ordered by inclusion of simplices. Then the identity map is a lattice isomorphism*

$$\mathrm{Sub}(K) \;\cong\; \mathrm{Down}(\mathcal{F}(K)),$$

onto the lattice of down-sets of $\mathcal{F}(K)$. Consequently $\mathrm{Sub}(K)$ is a finite distributive lattice, hence a finite Heyting algebra, its join-irreducible elements are exactly the closed simplices $\downarrow\sigma = \{\tau \in K \mid \tau \subseteq \sigma\}$ for $\sigma \in K$, and $J(\mathrm{Sub}(K)) \cong \mathcal{F}(K)$ as posets.

Proof. A set $A \subseteq K$ is face-closed exactly when it is down-closed in $\mathcal{F}(K)$, so $\mathrm{Sub}(K)$ and $\mathrm{Down}(\mathcal{F}(K))$ have the same underlying set; the identity map preserves $\subseteq$, intersections and unions in both directions and is therefore a lattice isomorphism. Down-set lattices are distributive, and every finite distributive lattice is a Heyting algebra. By Birkhoff's representation theorem, for a finite poset P the join-irreducibles of $\mathrm{Down}(P)$ are exactly the principal down-sets $\downarrow x$, $x \in P$, and $x \mapsto \downarrow x$ is an order isomorphism $P \cong J(\mathrm{Down}(P))$. □

Remark 2.16 (Alexandrov reading)**.** Theorem 2.15 identifies $\mathrm{Sub}(K)$ with the lattice of open sets of the Alexandrov topology on $\mathcal{F}(K)$ ordered by *reverse* inclusion: the down-sets of $\mathcal{F}(K)$ are precisely the sets open in that topology. Algebraically, therefore, SC-semantics is the Alexandrov semantics of a finite poset. This is the structural reason for the inertness phenomenon of Theorem 4.20: the propositional theory of $\mathrm{Sub}(K)$ sees nothing beyond the order type of $\mathcal{F}(K)$, and no topological invariant of K survives that passage. The value of the correspondence is not that it is unexpected but that it locates exactly where the geometric information is lost, which is what makes the modal enrichment of Section 9 necessary rather than optional.

Corollary 2.17 (Which finite Heyting algebras arise as $\mathrm{Sub}(K)$)**.** *Let* $\mathbf{H}$ *be a finite Heyting algebra and* $J(\mathbf{H})$ *the poset of its join-irreducible elements. Then* $\mathbf{H} \cong \mathrm{Sub}(K)$ *for some finite simplicial complex* K *if and only if* $J(\mathbf{H}) \cong \mathcal{F}(K)$ *for some such* K*. The latter condition is intrinsic: for a finite poset* P*, writing* $\mathcal{B}(X)$ *for the poset of non-empty subsets of a finite set* X *and* $\mathrm{At}(x) := \{\, y \in P \mid y \leq x,\ y \text{ minimal in } P \,\}$*, one has* $P \cong \mathcal{F}(K)$ *for some finite simplicial complex* K *if and only if*

1. (local simplex condition) *for every* $x \in P$ *there is a finite set* X_x *with* $\downarrow x \cong \mathcal{B}(X_x)$ *(equivalently,* P *is a* simplicial poset*), and*
2. (vertex-set separation) *the map* $x \mapsto \mathrm{At}(x)$ *is injective.*

In particular not every finite Heyting algebra is realisable: for the three-element chain $0 < a < 1$ *we have* $J(\mathbf{H}) = \{a, 1\}$*, a two-element chain, while* $|\mathcal{B}(X)| = 2^{|X|} - 1$ *is odd for every finite* X *and hence never equal to* 2*; condition (1) fails, so no complex satisfies* $|\mathrm{Sub}(K)| = 3$*.*

Proof. Every finite Heyting algebra is a finite distributive lattice, so Birkhoff's theorem gives $\mathbf{H} \cong \mathrm{Down}(J(\mathbf{H}))$; since $\mathrm{Down}(P) \cong \mathrm{Down}(Q)$ iff $P \cong Q$ for finite posets, Theorem 2.15 yields the first equivalence.

For the intrinsic description, suppose $P = \mathcal{F}(K)$. The down-set of a simplex σ is the poset of its non-empty subsets, giving (1) with $X_\sigma = \sigma$; the minimal elements below σ are its vertices and a simplex is determined by its vertex set, giving (2).

Conversely assume (1) and (2). Let V be the set of minimal elements of P and put $K := \{\mathrm{At}(x) \mid x \in P\}$, a collection of non-empty subsets of V since P is finite. Fix $x \in P$ and an isomorphism $\downarrow x \cong \mathcal{B}(X_x)$. The minimal elements of $\mathcal{B}(X_x)$ are its singletons, so the isomorphism carries X_x bijectively onto $\mathrm{At}(x)$ and, for every non-empty $T \subseteq \mathrm{At}(x)$, produces some $y \leq x$ with $\mathrm{At}(y) = T$. Hence K is closed under non-empty faces and is a simplicial complex over V. The map $x \mapsto \mathrm{At}(x)$ is onto K by construction and injective by (2), and it is an order isomorphism: $y \leq x$ gives $\mathrm{At}(y) \subseteq \mathrm{At}(x)$, and conversely if $\mathrm{At}(y) \subseteq \mathrm{At}(x)$ then the element of $\downarrow x$ with atom set $\mathrm{At}(y)$ coincides with y by (2). Thus $P \cong \mathcal{F}(K)$. □

Remark 2.18 (Condition (2) is not redundant)**.** Let P consist of two minimal elements u, v together with two distinct elements $e_1 \neq e_2$, each covering exactly u and v. Then $\downarrow e_i \cong \mathcal{B}(\{1,2\})$ and $\downarrow u \cong \downarrow v \cong \mathcal{B}(\{1\})$, so P is a simplicial poset and condition (1) holds. But $\mathrm{At}(e_1) = \mathrm{At}(e_2) = \{u, v\}$, and in a simplicial complex a simplex is determined by its vertex set, so P is not a face poset. Dropping condition (2) would therefore make Corollary 2.17 false.

Corollary 2.19 (The algebra is Boolean exactly in dimension zero)**.** *For a finite simplicial complex K the following are equivalent:*

1. $\mathrm{Sub}(K)$ *is a Boolean algebra;*
2. $\neg_H A = K \setminus A$ *for every* $A \in \mathrm{Sub}(K)$*;*
3. $\mathcal{F}(K)$ *is an antichain;*
4. $\dim K = 0$*, i.e. K is a finite set of isolated vertices.*

Thus a single edge already destroys Booleanity: the intuitionistic character of $\mathrm{Sub}(K)$ *is carried by the incidence relation, not by any global topological invariant of K.*

Proof. By Theorem 2.15, $\mathrm{Sub}(K) \cong \mathrm{Down}(\mathcal{F}(K))$. A finite down-set lattice $\mathrm{Down}(P)$ is Boolean iff P is an antichain: if $x < y$ in P, then any down-set disjoint from $\downarrow x$ omits y as well, so

$\downarrow x$ has no complement; conversely on an antichain every subset is a down-set and $\mathrm{Down}(P) = \mathcal{P}(P)$. This gives (1) $\Leftrightarrow$ (3), and (3) $\Leftrightarrow$ (4) because $\mathcal{F}(K)$ contains a strict inclusion iff K contains a simplex of dimension ≥ 1. If (4) holds, every subcomplex is a set of vertices and its complement is face-closed, so (2) follows from Corollary 2.13. If (4) fails, pick an edge $\{u, v\} \in K$ and set $A = \{\{u\}\}$; then $\{u, v\} \in K \setminus A$ while $\{u\} \notin K \setminus A$, so $K \setminus A$ is not face-closed and (2) fails. □

Remark 2.20 (A face-poset consequence)**.** While the representation of finite Heyting algebras as algebras of open sets of finite posets is classical (Esakia duality [12]; Birkhoff's representation of distributive lattices), the fact that $|\mathrm{Sub}(K)| = 3$ is *impossible* for any finite simplicial complex K is a specialization of that correspondence to the face-poset setting. No finite simplicial complex has exactly three subcomplexes: the subcomplexes of K always include $\emptyset$, every vertex (as a 0-simplex), and K itself, and the lattice structure is constrained by the face relation. This shows that "spatial" representation by simplicial complexes is strictly more restrictive than poset representation: not every finite Heyting algebra arises as $\mathrm{Sub}(K)$ for some K, even though every finite Heyting algebra arises as the open-set algebra of some finite poset.

The face poset order is on simplices (elements of $\mathcal{F}(K)$), whereas the subcomplexes correspond to the *ideals* (down-sets) of this poset. This distinction removes any ambiguity about what is ordered: the poset orders simplices by inclusion in faces, while the Heyting algebra $\mathrm{Sub}(K)$ is the lattice of down-sets of the face poset. The bridge between spatial semantics and the algebraic properties of SC-Logic is precisely this identification of formulas with ideals of the face poset, rather than with arbitrary subsets.

2.4. Valuations and the algebraic complement

Valuations are required to land in $\mathrm{Sub}(K)$ so that the logical operations are interpreted geometrically rather than merely set-theoretically. That constraint is what forces the interior repair of Remark 2.9, and hence the intuitionistic behaviour of the entire system.

Definition 2.21 (SC-valuation)**.** Let K be a finite simplicial complex. An *SC-valuation* on K is a map

$$v : \mathrm{Prop} \longrightarrow \mathrm{Sub}(K),$$

and the pair (K, v) is an *SC-model*. For $p \in \mathrm{Prop}$ we write $K_p := v(p)$ and call it the *truth set* of p. A valuation extends uniquely to all of $\mathcal{L}_0$ by

$$[\![p]\!] = K_p, \quad [\![\bot]\!] = 0, \quad [\![\varphi \wedge \psi]\!] = [\![\varphi]\!] \cap [\![\psi]\!], \quad [\![\varphi \vee \psi]\!] = [\![\varphi]\!] \cup [\![\psi]\!], \quad [\![\varphi \to \psi]\!] = [\![\varphi]\!] \to_H [\![\psi]\!],$$

which is well defined by Proposition 2.11; in particular $[\![\neg\varphi]\!] = \neg_H [\![\varphi]\!]$.

Definition 2.22 (Algebraic complement)**.** For $A \in \mathrm{Sub}(K)$, the *algebraic complement* is the plain set difference

$$\neg_{\mathrm{alg}} A := K \setminus A.$$

Remark 2.23. The operation $\neg_{\mathrm{alg}}$ is *not* internal to $\mathrm{Sub}(K)$: by Corollary 2.13 it returns a subcomplex exactly when A is closed under cofaces, and by Corollary 2.19 it fails to do so as soon as K contains a single edge. The Heyting complement is the largest face-closed part of the naive complement, so negation in $\mathrm{Sub}(K)$ is not set difference but its geometric repair, and Example 2.14 shows the repair can be drastic. Accordingly, $\neg_{\mathrm{alg}}$ is used in this paper only as a reference point against which the geometric content of $\neg_H$ is measured; it is never a connective of $\mathcal{L}_0$ or $\mathcal{L}_\Box$. The systematic study of the gap $\neg_{\mathrm{alg}} A \setminus \neg_H A$ is the subject of Section 6.

3. Kripke Semantics via Filtrations

The algebraic semantics of Section 2 answers the question *what* the connectives of SC-Logic mean, but not *how* truth is discovered. In practice a simplicial complex is rarely given at once: it is revealed in stages, as in a Čech or Vietoris–Rips filtration, where simplices appear as a scale parameter grows. Our aim in this section is to turn this observation into a semantics: the stages of a filtration will play the role of Kripke worlds, and the monotone growth of the complex will play the role of the accessibility relation. The heredity lemma below (Lemma 3.6) confirms that this reading is coherent, and Proposition 3.8 shows that it recovers exactly the algebraic semantics of Section 2, so nothing is lost, and a dynamic picture of truth is gained.

3.1. Filtration and the Intuitionistic Order

Definition 3.1. A *filtration* $F = (K_t)_{t \in T}$ is a monotone family of subcomplexes indexed by a poset $(T, \leq)$: whenever $s \leq t$ in T, we have $K_s \subseteq K_t$.

Remark 3.2. Each K_t carries a Heyting algebra structure by Proposition 2.11. The inclusions $K_s \hookrightarrow K_t$ for $s \leq t$ are *not* Heyting homomorphisms: they preserve neither the top element (the top of $\mathrm{Sub}(K_s)$ is $K_s \neq K_t$ whenever the inclusion is proper) nor negation. The natural structure map runs in the opposite direction, as restriction

$$\rho_{ts} : \mathrm{Sub}(K_t) \to \mathrm{Sub}(K_s), \qquad A \mapsto A \cap K_s,$$

so F is an *inverse system* of Heyting algebras over $(T, \leq)$, equivalently a presheaf, rather than a directed system.

3.2. The Filtration Frame

Definition 3.3 (Filtration Frame)**.** Given a filtration $F = (K_t)_{t \in T}$ (Definition 3.1), the associated *Kripke frame* is $(T, \leq)$, where each index $t \in T$ is a *world* and the accessibility relation is the partial order $\leq$. The intuitionistic reading is that worlds are *stages of construction* of the complex, and $s \leq t$ means that stage t refines (extends) stage s: every simplex present at stage s persists at stage t, and new simplices may appear.

Definition 3.4 (Persistent Valuation)**.** A *valuation* V assigns to each atomic proposition p and each world t a subcomplex $\mathrm{val}(p, t) \in \mathrm{Sub}(K_t)$, subject to the *persistence condition*:

$$s \leq t \implies \mathrm{val}(p, s) \subseteq \mathrm{val}(p, t).$$

This is exactly the heredity (monotonicity) condition of intuitionistic Kripke models: once a proposition becomes true at a stage, it remains true at all later stages.

3.3. The Forcing Relation

With frames and valuations in place, we now define forcing so that each truth set is face-closed by construction. Intersections and unions of subcomplexes are automatically face-closed, so conjunction and disjunction need no repair; the operator int_{K_t} is applied only in the implication, negation, and modal clauses, whose defining conditions need not be face-closed on their own. We write $[\![\varphi]\!]_t \subseteq K_t$ for the truth set of φ at stage t.

Definition 3.5 (Forcing)**.** Fix a filtration frame $(T, \leq)$ and a persistent valuation V. For a world $t \in T$, a simplex $\sigma \in K_t$, and a formula φ, define the truth sets $[\![\varphi]\!]_t \subseteq K_t$ by induction on φ:

1. $[\![p]\!]_t = \mathrm{val}(p,t)$, for each atomic proposition p.
2. $[\![\varphi \wedge \psi]\!]_t = [\![\varphi]\!]_t \cap [\![\psi]\!]_t$.
3. $[\![\varphi \vee \psi]\!]_t = [\![\varphi]\!]_t \cup [\![\psi]\!]_t$.
4. $[\![\varphi \to \psi]\!]_t = \mathrm{int}_{K_t}\Big(\{\sigma \in K_t \mid \forall t' \geq t,\ \sigma \in K_{t'} \Rightarrow \sigma \notin [\![\varphi]\!]_{t'} \text{ or } \sigma \in [\![\psi]\!]_{t'}\}\Big)$.
5. $[\![\neg\varphi]\!]_t = \mathrm{int}_{K_t}\Big(\{\sigma \in K_t \mid \forall t' \geq t,\ \sigma \in K_{t'} \Rightarrow \sigma \notin [\![\varphi]\!]_{t'}\}\Big)$.
6. $[\![\bot]\!]_t = \emptyset$.
7. $[\![\Box\varphi]\!]_t = \mathrm{int}_{K_t}\big(B_t(\varphi)\big)$, where $B_t(\varphi) = \{\sigma \in K_t \mid \forall t' \geq t,\ \forall \tau \supseteq \sigma \text{ in } K_{t'},\ \tau \in [\![\varphi]\!]_{t'}\}$.

$$t \Vdash_\sigma \varphi \quad \Longleftrightarrow \quad \sigma \in [\![\varphi]\!]_t.$$

Lemma 3.6 (Monotonicity/Heredity)**.** *If $t \Vdash_\sigma \varphi$ and $t \leq t'$ with $\sigma \in K_{t'}$, then $t' \Vdash_\sigma \varphi$.*

Proof. By induction on φ. Atomic and $\bot$ cases use $\mathrm{val}(p,t) \subseteq \mathrm{val}(p,t')$ (Definition 3.4) and emptiness, respectively. For conjunction, intersection is monotone. For disjunction, the union of two subcomplexes is face-closed in $\mathrm{Sub}(K_{t'})$; persistence and the induction hypothesis give inclusion at t'. For implication, suppose $t \Vdash_\sigma \varphi \to \psi$, i.e. $\sigma \in [\![\varphi \to \psi]\!]_t = \mathrm{int}_{K_t}(S_t)$, where

$$S_t = \{\tau \in K_t : \forall u \geq t,\ \tau \in K_u \Rightarrow (\tau \notin [\![\varphi]\!]_u \text{ or } \tau \in [\![\psi]\!]_u)\}.$$

The defining condition on S_t quantifies over all $u \geq t$. Since $t \leq t'$, every $u \geq t'$ also satisfies $u \geq t$, so the same condition holds and $S_t \cap K_{t'} \subseteq S_{t'}$. As $\sigma \in \mathrm{int}_{K_t}(S_t)$ there is a face-open neighbourhood $U \ni \sigma$ in K_t with $U \subseteq S_t$. Since U is face-closed in K_t and $K_t \subseteq K_{t'}$, it is also face-closed in $K_{t'}$; thus openness is preserved when passing to $K_{t'}$. By persistence $U \subseteq K_t \subseteq K_{t'}$, and the defining condition above gives $U \subseteq S_{t'}$. Therefore $\sigma \in \mathrm{int}_{K_{t'}}(S_{t'}) = [\![\varphi \to \psi]\!]_{t'}$. Negation is the special case $\psi = \bot$.

□Informally, Lemma 3.6 says that in a filtration truth is *persistent*: once a formula is established at some scale, it survives at all coarser scales. This is the logical shadow of persistence in topological data analysis, and it is precisely the property that makes the intuitionistic reading of SC-Logic natural rather than imposed.

3.4. Agreement of Algebraic and Kripke Semantics

Lemma 3.7 (Truth Sets are Subcomplexes)**.** *For every world t, $[\![\varphi]\!]_t$ is a subcomplex of K_t for every formula φ.*

Proof. By induction on φ. Atoms are subcomplexes by Definition 3.4, and $\emptyset$ is a subcomplex. Intersections and unions of subcomplexes are subcomplexes, and int_{K_t} returns a subcomplex by Lemma 2.6; the implication, negation, and modal clauses are each of the form $\mathrm{int}_{K_t}(S)$. □

Proposition 3.8 (Agreement of Algebraic and Kripke Semantics)**.** *Restrict to a single world t with the discrete filtration ($T = \{t\}$, $K_t = K$). Then, under Lemma 3.7, the map $\varphi \mapsto [\![\varphi]\!]_t$ computes exactly the Heyting operations of Proposition 2.11:*

$$[\![\varphi\wedge\psi]\!]_t = [\![\varphi]\!]_t \cap [\![\psi]\!]_t, \quad [\![\varphi\vee\psi]\!]_t = [\![\varphi]\!]_t \cup [\![\psi]\!]_t, \quad [\![\varphi \to \psi]\!]_t = [\![\varphi]\!]_t \to_H [\![\psi]\!]_t, \quad [\![\bot]\!]_t = \emptyset.$$

Proof. With a discrete filtration the unique world t satisfies $t' \geq t \Rightarrow t' = t$, so the implication clause of Definition 3.5 collapses to

$$[\![\varphi \to \psi]\!]_t = \mathrm{int}_K\big((K \setminus [\![\varphi]\!]_t) \cup [\![\psi]\!]_t\big),$$

which is Definition 2.8; taking $\psi = \bot$ gives $[\![\neg\varphi]\!]_t = \mathrm{int}_K(K \setminus [\![\varphi]\!]_t)$. Meet and join are the lattice operations of $\mathrm{Sub}(K)$, since $\mathrm{Sub}(K)$ is closed under intersection and union by Lemma 2.6. □

Remark 3.9 (Scope of the agreement result)**.** Proposition 3.8 is deliberately restricted to the discrete filtration. On a nontrivial filtration the two semantics are *not* freely interchangeable: the forcing clause for implication quantifies over future stages, whereas $\to_H$ is computed statically on $\mathrm{Sub}(K)$, and the two diverge as soon as an atom is empty at an early stage and non-empty later (Example 3.10). What survives, and all that the completeness proof of Section 4 uses, is that the canonical forcing model is a Kripke *representation* of the Lindenbaum–Tarski algebra via prime filters, not a formula-by-formula identity on arbitrary filtrations. For filtrations indexed by a dense parameter both readings remain well defined, but we do not know whether they agree; we record this as an open problem, since no result in Sections 3–6 depends on the dense case.

Example 3.10. Let $T = \{0, 1, 2\}$ with the usual order, and let $K = \partial\Delta^2$ be as in Example 2.5, with the delayed-appearance filtration: $K_0 = \{\{a\}, \{b\}, \{c\}\}$, $K_1 = K_0 \cup \{\{a, b\}\}$, $K_2 = \partial\Delta^2$. Take a single atom p with $\mathrm{val}(p, 0) =$

$\mathrm{val}(p,1) = \emptyset$ and $\mathrm{val}(p,2) = \{\{c\}\}$; this valuation is persistent and face-hereditary. At stage 0 we have $[\![p]\!]_0 = \emptyset$, hence $0 \nVdash_{\{c\}} p$. Moreover $\{c\} \in K_2$ and $\{c\} \in [\![p]\!]_2$, so the defining condition of the negation clause fails at $\{c\}$ and $0 \nVdash_{\{c\}} \neg p$. Since $[\![p \vee \neg p]\!]_0 = \emptyset \cup \{\{a\},\{b\}\} = \{\{a\},\{b\}\}$, we get $0 \nVdash_{\{c\}} p \vee \neg p$: the law of excluded middle fails, because the future stage 2 can still make p true at $\{c\}$.

4. Soundness and Completeness

4.1. The SC-Logic Proof System

Recall the language $\mathcal{L}_\Box$ fixed in Section 1.

Definition 4.1 (SC-Logic: Axioms and Rule)**.** The axiom schemata of SC-Logic are those of intuitionistic propositional calculus:

$$
\begin{aligned}
&A_1 : \varphi \to (\psi \to \varphi)\\
&A_2 : (\varphi \to (\psi \to \chi)) \to ((\varphi \to \psi) \to (\varphi \to \chi))\\
&A_3 : \varphi \to (\psi \to (\varphi \wedge \psi))\\
&A_4 : (\varphi \wedge \psi) \to \varphi\\
&A_5 : (\varphi \wedge \psi) \to \psi\\
&A_6 : \varphi \to (\varphi \vee \psi)\\
&A_7 : \psi \to (\varphi \vee \psi)\\
&A_8 : (\varphi \to \chi) \to ((\psi \to \chi) \to ((\varphi \vee \psi) \to \chi))\\
&A_9 : \bot \to \varphi
\end{aligned}
$$

The sole inference rule is modus ponens: from φ and $\varphi \to \psi$ infer ψ. No semantic condition is part of this definition.

Definition 4.2 (Derivability)**.** Let Γ be a set of formulas. We write $\Gamma \vdash \varphi$ if there is a finite sequence $\varphi_1, \ldots, \varphi_n = \varphi$ in which each φ_i is an instance of an axiom schema of Definition 4.1, an element of Γ, or follows from φ_j and $\varphi_k = \varphi_j \to \varphi_i$ with $j, k < i$ by modus ponens. This relation is purely syntactic.

Lemma 4.3 (Definability of Negation)**.** *For every formula φ, every filtration model, every world t and every simplex $\sigma \in K_t$,*

$$t \Vdash_\sigma \neg\varphi \quad \Longleftrightarrow \quad t \Vdash_\sigma \varphi \to \bot.$$

Hence clause (4) of Definition 3.5 is redundant but consistent.

Proof. Since $[\![\bot]\!]_{t'} = \emptyset$ for every t', the implication clause of Definition 3.5 instantiated at $\psi = \bot$ becomes

$$[\![\varphi \to \bot]\!]_t = \mathrm{int}\Big(\{\sigma \in K_t \mid \forall t' \geq t,\ \sigma \in K_{t'} \Rightarrow \sigma \notin [\![\varphi]\!]_{t'}\}\Big),$$

which is literally the negation clause. The two truth sets therefore coincide. □

4.2. Soundness

Definition 4.4 (Validity)**.** A formula φ is *valid* in a filtration model $(T, \leq, V)$ if $[\![\varphi]\!]_t = K_t$ for every world $t \in T$, and *SC-valid*, written $\vDash \varphi$, if it is valid in every filtration model. More generally, $\Gamma \vDash \varphi$ means that every filtration model $(T, \leq, V)$, world t and simplex $\sigma \in K_t$ with $t \Vdash_\sigma \gamma$ for all $\gamma \in \Gamma$ also satisfies $t \Vdash_\sigma \varphi$.

Theorem 4.5 (Soundness)**.** *If* $\Gamma \vdash \varphi$ *then* $\Gamma \vDash \varphi$.

Proof. By induction on the length of the derivation. For the base case we check that each axiom schema is forced at every world and simplex; we verify the implication and negation axioms explicitly using Definition 3.5.

Axiom A_1, $\varphi \to (\psi \to \varphi)$. Fix (t, σ). By the implication clause we must show that for every $t' \geq t$ with $\sigma \in K_{t'}$ and $t' \Vdash_\sigma \varphi$ one has $t' \Vdash_\sigma \psi \to \varphi$; and for that, that every $t'' \geq t'$ with $t'' \Vdash_\sigma \psi$ satisfies $t'' \Vdash_\sigma \varphi$. Fix such t''. Since $t'' \geq t'$ and $t' \Vdash_\sigma \varphi$, persistence (Lemma 3.6) gives $t'' \Vdash_\sigma \varphi$, as required.

Axiom A_9 *($\bot \to \varphi$):* For every $t' \geq t$ with $\sigma \in K_{t'}$ we have $\sigma \notin [\![\bot]\!]_{t'} = \emptyset$ vacuously, so the implication clause of Definition 3.5 yields $t \Vdash_\sigma \bot \to \varphi$ at every (t, σ).

For completeness, the remaining axiom schemata are checked explicitly; they are similar because the implication clause quantifies over every $t' \geq t$. A_2 is the distributive composition of implications $(\varphi \to (\psi \to \chi)) \to ((\varphi \to \psi) \to (\varphi \to \chi))$; the implication clause quantifies over the same future stages as the Kripke clause, and Lemma 3.6 supplies the required persistence. A_3 is $\varphi \to (\psi \to (\varphi \wedge \psi))$, using the intersection clause for $\wedge$; A_4 and A_5 are the two projections $(\varphi \wedge \psi) \to \varphi$ and $(\varphi \wedge \psi) \to \psi$, again by intersection. A_6 and A_7 are $\varphi \to (\varphi \vee \psi)$ and $\psi \to (\varphi \vee \psi)$, using the union clause for $\vee$. A_8, $(\varphi \to \chi) \to ((\psi \to \chi) \to ((\varphi \vee \psi) \to \chi))$, follows from the same union and future-stage implication clauses. In each verification, the relevant formulas lie in $\mathsf{Sub}(\varphi)$, persistence is provided by Lemma 3.6, and the implication check includes the current stage $t' = t$. There are no separate modal axiom

schemata in this system: these nine schemata A_1–A_9, together with modus ponens, are exactly the stated intuitionistic axiomatization. Finally, for the inductive modus-ponens step, if $\Gamma \vDash \psi$ and $\Gamma \vDash \psi \to \varphi$, then any model of Γ forces both at every (t, σ); taking $t' = t$ in the implication clause yields $t \Vdash_\sigma \varphi$. □

The verification is uniform for a reason: heredity of the forcing relation under passage to later stages, together with closure under faces, already fixes the behaviour of each connective, so the individual axiom schemata require no separate idea.

Lemma 4.6 (Filtration realization)**.** *Let $(P, \leq)$ be a finite poset and let $K = \Delta(P)$ be its order complex. For $t \in P$ set $K_t = \Delta(\downarrow t)$, where $\downarrow t = \{s : s \leq t\}$. Then:*

1. *each K_t is a nonempty subcomplex of the finite complex K;*
2. *$s \leq t \Rightarrow K_s \subseteq K_t$ (monotonicity); and*
3. *$K_s \subseteq K_t \Rightarrow s \leq t$ (order reflection).*

Hence $t \mapsto K_t$ is an order isomorphism of $(P, \leq)$ onto $(\{K_t\}, \subseteq)$.

Proof. (1) P is finite, so $K = \Delta(P)$ is a finite complex and $K_t = \Delta(\downarrow t) \subseteq K$ is a subcomplex; it is nonempty since $t \in \downarrow t$ yields the vertex $\{t\} \in K_t$.

(2) If $s \leq t$ then $\downarrow s \subseteq \downarrow t$, so every chain of $\downarrow s$ is a chain of $\downarrow t$, i.e. $K_s \subseteq K_t$.

(3) If $K_s \subseteq K_t$, then $\{s\} \in K_s \subseteq K_t = \Delta(\downarrow t)$, so $s \in \downarrow t$, that is $s \leq t$.

Combining (2) and (3), $s \leq t \iff K_s \subseteq K_t$. □

Remark 4.7. Monotonicity of a filtration (condition (2), a property of the *complexes* K_t) and persistence of a valuation $s \leq t \Rightarrow \mathrm{val}(p, s) \subseteq \mathrm{val}(p, t)$, a property of the *valuation*) are logically independent; neither implies the other, and both are imposed separately in Definition 3.1. The realization above supplies only the geometric side, and any valuation placed on (K_t) must be checked for persistence in its own right.

4.3. Completeness

Theorem 4.8 (Strong Completeness for $\mathcal{L}_0$)**.** *For all $\Gamma \cup \{\varphi\} \subseteq \mathcal{L}_0$, if $\Gamma \vDash \varphi$ then $\Gamma \vdash \varphi$.*

Remark 4.9. Completeness is asserted for the $\Box$-free fragment $\mathcal{L}_0$, matching the intuitionistic axiomatization of Definition 4.1. A completeness result for the full $\mathcal{L}_\Box$ would require an explicit modal axiomatization of $\Box$ and is outside the present scope; the semantic behavior of $\Box$ under non-degeneracy is instead recorded by Theorem 4.17.

Definition 4.10 (Point Filtration)**.** Let $(W, \leq)$ be a poset and let $\Vdash^K$ be an intuitionistic Kripke valuation on it, i.e. the atomic assignments are upward closed under $\leq$. Fix the one-point complex $\mathbf{1} = \{\{x\}\}$. Define $K_w := \mathbf{1}$ for every $w \in W$, and

$$\mathrm{val}(p, w) := \begin{cases} \{x\} & \text{if } w \Vdash^K p, \\ \emptyset & \text{otherwise.} \end{cases}$$

Lemma 4.11 (Legitimacy of the point filtration)**.** *For any poset $(W, \leq)$ and any Kripke valuation $\Vdash^K$, the point filtration of Definition 4.10 is a filtration model in the sense of Definition 3.1: $(K_w)_{w \in W}$ is a monotone family of nonempty subcomplexes of $\mathbf{1}$, and $\mathrm{val}(p, \cdot)$ is a persistent valuation. No finiteness of W is required.*

Proof. Every stage is $K_w = \mathbf{1} = \{\{x\}\}$, so $K_w \neq \emptyset$ and $K_w \in \mathrm{Sub}(\mathbf{1})$. For $s \leq t$, $K_s = \mathbf{1} = K_t$, hence $K_s \subseteq K_t$: the family is monotone (indeed constant). For persistence, if $s \leq t$ and $\mathrm{val}(p, s) = \{x\}$, then $s \Vdash^K p$ by Definition 4.10, and upward persistence of Kripke valuations gives $t \Vdash^K p$, i.e. $\mathrm{val}(p, t) = \{x\}$; the only other case is $\mathrm{val}(p, s) = \emptyset \subseteq \mathrm{val}(p, t)$. No cardinality assumption on W or on any premise set is used. $\square$

Lemma 4.12 (Forcing Transfer)**.** *On a point filtration, for every $\Box$-free formula $\psi \in \mathcal{L}_0$ and every world w,*

$$w \Vdash_x \psi \quad \Longleftrightarrow \quad w \Vdash^K \psi,$$

where the right-hand side is standard intuitionistic Kripke forcing on $(W, \leq)$.

Proof. By induction on the structure of $\psi \in \mathcal{L}_0$; since $\mathcal{L}_0$ contains no $\Box$, the connectives to treat are exactly $p, \bot, \wedge, \vee, \to$ (with $\neg$ an abbreviation). Every stage carries the single simplex $\{x\}$, so $K_w = \mathbf{1} = \{\{x\}\}$; the operators cl_{K_w} and int_{K_w} are the identity on $\mathrm{Sub}(\mathbf{1}) = \{\emptyset, \mathbf{1}\}$, and a subcomplex is nonempty iff it equals $\mathbf{1}$.

Atoms. By Definition 4.10, $\{x\} \in [\![p]\!]_w \iff \mathrm{val}(p, w) = \{x\} \iff w \Vdash^K p$.

$\bot$. $[\![\bot]\!]_w = \emptyset$, so $\{x\} \notin [\![\bot]\!]_w$ and $w \nVdash^K \bot$.

$\wedge, \vee$. Intersection and union of subcomplexes of $\mathbf{1}$ correspond to conjunction and disjunction of the truth values $\{x\} \in [\![\cdot]\!]_w$; apply the induction hypothesis.

$\to$. Since int_{K_w} is the identity, clause (4) of Definition 3.5 evaluated at $\sigma = \{x\}$ reads $\forall w' \geq w\,(\{x\} \in [\![\chi]\!]_{w'} \Rightarrow \{x\} \in [\![\theta]\!]_{w'})$, which by the induction hypothesis is $\forall w' \geq w\,(w' \Vdash^K \chi \Rightarrow w' \Vdash^K \theta)$, the Kripke clause for $\chi \to \theta$. Negation is the case $\theta = \bot$ via Lemma 4.3. □

Proof of Theorem 4.8. Suppose $\Gamma \nvdash \varphi$. The axioms and the single rule of Definition 4.1 are exactly those of IPC, so strong Kripke completeness for IPC [20] yields a poset $(W, \leq)$, a valuation $\Vdash^K$, and a world w_0 with $w_0 \Vdash^K \gamma$ for all $\gamma \in \Gamma$ and $w_0 \nVdash^K \varphi$. By Lemma 4.11 the associated point filtration is a legitimate filtration model, and by Lemma 4.12 it satisfies $w_0 \Vdash_x \gamma$ for all $\gamma \in \Gamma$ while $w_0 \nVdash_x \varphi$. Hence $\Gamma \not\models \varphi$. Contrapositively, $\Gamma \models \varphi$ implies $\Gamma \vdash \varphi$. No finiteness assumption on T or on Γ is used, so completeness is strong. □

Remark 4.13 (Algebraic reading of completeness)**.** The point-filtration argument above can equivalently be phrased algebraically: the Lindenbaum–Tarski algebra of SC-Logic is a Heyting algebra whose prime filters are exactly the worlds of the Kripke countermodel produced by IPC completeness. Realising each such world as the one-point complex $\mathbf{1}$ recovers the filtration model of Lemma 4.12; we therefore do not develop the canonical model separately.

Remark 4.14. Completeness is relative to the class of *filtration models*, Kripke frames arising from monotone families of subcomplexes, rather than arbitrary Kripke frames. The point-filtration construction shows that this class already captures full intuitionistic provability, so restricting truth to geometric witnesses loses no logical power.

4.4. Decidability and Finite Model Property

Proposition 4.15 (Finite Model Property)**.** *If φ is not a theorem of SC-Logic, there is a finite countermodel that is a point filtration over the fixed ambient one-point complex $\mathbf{1} = \{\{x\}\}$, falsifying φ, with the bound*

$$N := 2^{|\mathrm{Sub}(\varphi)|}, \qquad |W| \;\leq\; N$$

on the number of stages (worlds), where $\mathrm{Sub}(\varphi)$ *denotes the set of subformulas of φ.*

Proof. By Theorem 4.8, a non-theorem φ is falsified in a point-filtration model arising from an intuitionistic Kripke countermodel $(W, \leq, \Vdash^K)$. We filter this model by subformulas.

Put $\Sigma = \mathrm{Sub}(\varphi)$, closed under subformulas. Define a preorder on W by

$$w \preceq_\Sigma w' \iff \forall \psi \in \Sigma\, (w \Vdash^K \psi \Rightarrow w' \Vdash^K \psi).$$

Reflexivity is immediate because $w \Vdash^K \psi \Rightarrow w \Vdash^K \psi$; transitivity follows because if $w \preceq_\Sigma w'$ and $w' \preceq_\Sigma w''$ then for every $\psi \in \Sigma$, $w \Vdash^K \psi \Rightarrow w' \Vdash^K \psi \Rightarrow w'' \Vdash^K \psi$, so $w \preceq_\Sigma w''$. Define

$$w \sim w' \iff w \preceq_\Sigma w' \text{ and } w' \preceq_\Sigma w.$$

This is an equivalence relation: reflexivity and symmetry are immediate from the definition of $\sim$, and transitivity follows because if $w \sim w'$ and $w' \sim w''$ then $w \preceq_\Sigma w'$ and $w' \preceq_\Sigma w''$, hence $w \preceq_\Sigma w''$ by transitivity of $\preceq_\Sigma$; symmetrically $w'' \preceq_\Sigma w$, so $w \sim w''$. Since Σ is finite, $\sim$ has at most $2^{|\Sigma|}$ classes, so $W/\!\sim$ is finite. Define the quotient relation by

$$[s] \leq^\sharp [t] \iff s \preceq_\Sigma t.$$

This is well defined. Suppose $s \sim s'$ and $t \sim t'$. If $[s] \leq^\sharp [t]$, i.e. $s \preceq_\Sigma t$, then from $s' \preceq_\Sigma s$ (as $s' \sim s$) and $t \preceq_\Sigma t'$ (as $t \sim t'$) we obtain $s' \preceq_\Sigma s \preceq_\Sigma t \preceq_\Sigma t'$, hence $s' \preceq_\Sigma t'$, i.e. $[s'] \leq^\sharp [t']$. Conversely, if $s' \preceq_\Sigma t'$, then from $s \preceq_\Sigma s'$ and $t' \preceq_\Sigma t$ we get $s \preceq_\Sigma s' \preceq_\Sigma t' \preceq_\Sigma t$, so $s \preceq_\Sigma t$. Hence $[s] \leq^\sharp [t]$ does not depend on the chosen representatives. It is reflexive because $\preceq_\Sigma$ is, transitive because $\preceq_\Sigma$ is, and antisymmetric because $[s] \leq^\sharp [t]$ and $[t] \leq^\sharp [s]$ give $s \preceq_\Sigma t$ and $t \preceq_\Sigma s$, i.e. $s \sim t$, so $[s] = [t]$. Thus $\leq^\sharp$ is a partial order.

We now verify the filtration conditions. (F1) If $w \leq w'$ in W and $\psi \in \Sigma$ with $w \Vdash^K \psi$, then $w' \Vdash^K \psi$ by the heredity of Kripke forcing, so $w \preceq_\Sigma w'$ and hence $[w] \leq^\sharp [w']$. (F2) If $[w] \leq^\sharp [w']$ and $\psi \in \Sigma$ with $w \Vdash^K \psi$, then $w \preceq_\Sigma w'$, so by definition $w' \Vdash^K \psi$.

Filtration Lemma. For all $\psi \in \Sigma$ and all $w \in W$,

$$w \Vdash^K \psi \iff [w] \Vdash^Q \psi,$$

where $[w] \Vdash^Q p \iff w \Vdash^K p$ for atoms and the forcing $\Vdash^Q$ on $W/\!\sim$ is defined by the standard Kripke clauses with respect to the frame $(W/\!\sim, \leq^\sharp)$. We prove this by induction on ψ.

Atoms. By definition $[w] \Vdash^Q p \iff w \Vdash^K p$.

Conjunction. $w \Vdash^K \psi \wedge \theta \iff w \Vdash^K \psi$ and $w \Vdash^K \theta$, which by the induction hypothesis is equivalent to $[w] \Vdash^Q \psi$ and $[w] \Vdash^Q \theta$, i.e. $[w] \Vdash^Q \psi \wedge \theta$.

Disjunction. $w \Vdash^K \psi \vee \theta \iff w \Vdash^K \psi$ or $w \Vdash^K \theta$, which by the induction hypothesis is equivalent to $[w] \Vdash^Q \psi$ or $[w] \Vdash^Q \theta$, i.e. $[w] \Vdash^Q \psi \vee \theta$.

Implication. Let $\psi \to \theta \in \Sigma$. For the left-to-right direction, suppose $w \Vdash^K \psi \to \theta$ and $[w] \leq^\sharp [w']$ with $[w'] \Vdash^Q \psi$. By the induction hypothesis $w' \Vdash^K \psi$. Since $[w] \leq^\sharp [w']$ we have $w \preceq_\Sigma w'$, and because $\psi \to \theta \in \Sigma$ and $w \Vdash^K \psi \to \theta$, the definition of $\preceq_\Sigma$ gives $w' \Vdash^K \psi \to \theta$. Together with $w' \Vdash^K \psi$ this yields $w' \Vdash^K \theta$, so by the induction hypothesis $[w'] \Vdash^Q \theta$. Hence $[w] \Vdash^Q \psi \to \theta$.

For the right-to-left direction we argue contrapositively. Suppose $w \nVdash^K \psi \to \theta$. Then there is a stage $w' \geq w$ in W with $w' \Vdash^K \psi$ and $w' \nVdash^K \theta$. By (F1), $[w] \leq^\sharp [w']$. By the induction hypothesis for ψ we have $[w'] \Vdash^Q \psi$, and by the induction hypothesis for θ we have $[w'] \nVdash^Q \theta$. Since $[w] \leq^\sharp [w']$ with $[w'] \Vdash^Q \psi$ and $[w'] \nVdash^Q \theta$, the Kripke implication clause for $\Vdash^Q$ gives $[w] \nVdash^Q \psi \to \theta$.

This completes the induction.

Finally, realise the finite frame $(W/\!\sim, \leq^\sharp, \Vdash^Q)$ as the point filtration over the fixed ambient complex $\mathbf{1}$, by setting $K_{[t]} = \mathbf{1}$ and $\mathrm{val}(p, [t]) = \{x\}$ iff $[t] \Vdash^Q p$. By Lemma 4.11 this is a filtration model and by Lemma 4.12 its forcing agrees with $\Vdash^Q$; combined with the Filtration Lemma it falsifies φ. The number of stages is $|W/\!\sim| \leq N$. □The bound $N = 2^{|\mathrm{Sub}(\varphi)|}$ is not claimed to be sharp. The point of the preorder $\preceq_\Sigma$ is that it collapses exactly the information φ cannot distinguish, so the quotient retains the finite combinatorial shadow of the original model and nothing more.

Corollary 4.16 (Decidability)**.** *SC-Logic is decidable.*

Proof. By Proposition 4.15, compute $N = 2^{|\mathrm{Sub}(\varphi)|}$. Enumerate, up to isomorphism, all finite posets $(W, \leq)$ with at most N elements. For each atom occurring in φ, enumerate every upward-closed subset of W, hence every persistent atomic valuation. For each resulting (poset, valuation) pair, build the point filtration model with $K_w = \mathbf{1}$, and model-check φ on that model. The search space is finite and effectively enumerable: there are finitely many posets of size at most N, finitely many atoms in φ, and finitely many upward-closed valuations for each poset. Consequently the algorithm terminates; it returns "not a theorem" exactly when a countermodel is found, and otherwise returns "theorem" by the finite model property. This is a decision procedure

in principle: the enumeration of all posets with at most N elements is far too coarse to be run, and no attempt is made here to optimize it. Its only role is to certify decidability by exhibiting a finite, effectively searchable space of countermodels. □

4.5. No-Go: The Logical Content of Geometric Non-Degeneracy

Theorem 4.17 (Two-Valued Stability)**.** *Let $(T, \leq)$ be any filtration with nonempty stages, i.e. $K_t \neq \emptyset$ for all t, and let V be a persistent valuation for which every atomic truth set is two-valued at each stage,* $\mathrm{val}(p, t) \in \{\emptyset, K_t\}$ *for all p, t. Then for every formula $\varphi \in \mathcal{L}_\Box$ and every stage t,*

$$[\![\varphi]\!]_t \in \{\emptyset, K_t\}.$$

Moreover the assignment $\varphi \mapsto ([\![\varphi]\!]_t = K_t)$ reproduces exactly standard intuitionistic S4*-Kripke forcing on $(T, \leq)$, with $\Box$ interpreted by the future-persistence clause of Definition 3.5.*

Proof. Because $K_t \neq \emptyset$, the values $\emptyset$ and K_t are distinct, so "$[\![\varphi]\!]_t = K_t$" is an unambiguous truth predicate; write $t \Vdash \varphi$ for it. On a nonempty stage both cl_{K_t} and int_{K_t} fix $\emptyset$ and K_t: $\mathrm{int}_{K_t}(K_t) = K_t$, $\mathrm{int}_{K_t}(\emptyset) = \emptyset$, and dually for closure. We induct on φ, verifying two-valuedness and agreement with the Kripke clause together.

Atoms. $\mathrm{val}(p, t) \in \{\emptyset, K_t\}$ by hypothesis, and $\mathrm{val}(p, t) = K_t$ is the Kripke atomic clause.

$\bot$. $[\![\bot]\!]_t = \emptyset \neq K_t$; no world forces $\bot$.

$\wedge$. $[\![\varphi \wedge \psi]\!]_t = [\![\varphi]\!]_t \cap [\![\psi]\!]_t$; by the induction hypothesis this is K_t iff both conjuncts are K_t.

$\vee$. Dually, $[\![\varphi \vee \psi]\!]_t = [\![\varphi]\!]_t \cup [\![\psi]\!]_t = K_t$ iff at least one disjunct is K_t.

$\to$. By Definition 3.5, $[\![\varphi \to \psi]\!]_t = \mathrm{int}_{K_t}\{\sigma : \forall t' \geq t,\ \sigma \in K_{t'} \text{ and } \sigma \in [\![\varphi]\!]_{t'} \Rightarrow \sigma \in [\![\psi]\!]_{t'}\}$. If for every $t' \geq t$ we have $t' \Vdash \varphi \Rightarrow t' \Vdash \psi$, the bracketed set is K_t and its interior is K_t. Otherwise some $t' \geq t$ has $[\![\varphi]\!]_{t'} = K_{t'} \neq \emptyset$ and $[\![\psi]\!]_{t'} = \emptyset$; then no $\sigma \in K_t \subseteq K_{t'}$ survives, the set is $\emptyset$, and its interior is $\emptyset$. This is the Kripke clause $t \Vdash \varphi \to \psi \iff \forall t' \geq t\,(t' \Vdash \varphi \Rightarrow t' \Vdash \psi)$. Negation is the instance $\psi = \bot$ (Lemma 4.3).

$\Box$. By Definition 3.5, $[\![\Box\varphi]\!]_t = \mathrm{int}_{K_t}(B_t(\varphi))$, where $B_t(\varphi) = \{\sigma \in K_t : \forall t' \geq t,\ \forall \tau \supseteq \sigma \text{ in } K_{t'},\ \tau \in [\![\varphi]\!]_{t'}\}$. By the induction hypothesis each $[\![\varphi]\!]_{t'} \in \{\emptyset, K_{t'}\}$. If $t' \Vdash \varphi$ for all $t' \geq t$, then $[\![\varphi]\!]_{t'} = K_{t'}$ contains every $\tau \in K_{t'}$, so $B_t(\varphi) = K_t$ and its interior is K_t. Otherwise some $t' \geq t$ has $[\![\varphi]\!]_{t'} = \emptyset$; taking $\tau = \sigma \in K_t \subseteq K_{t'}$ shows no σ lies in $B_t(\varphi)$, so $B_t(\varphi) = \emptyset$ and

its interior is $\emptyset$. Hence $\Box\varphi$ is two-valued and $t \Vdash \Box\varphi \iff \forall t' \geq t\,(t' \Vdash \varphi)$, the $\mathsf{S4}$ clause.

In every case $[\![\varphi]\!]_t \in \{\emptyset, K_t\}$ and the induced truth predicate obeys the corresponding Kripke clause, completing the induction. □

Lemma 4.18 (Geometric inflation)**.** *Let $(T, \leq)$ be finite, let $(K_t)_{t\in T}$ and $(K'_t)_{t\in T}$ be monotone filtrations with nonempty stages and $K_t \subseteq K'_t$ for all $t \in T$, and let V be a persistent two-valued valuation on $(K_t)_{t\in T}$. Define $\mathrm{val}'(p,t) = K'_t$ whenever $\mathrm{val}(p,t) = K_t$, and $\mathrm{val}'(p,t) = \emptyset$ otherwise. Then V' is persistent and two-valued, and for every $\varphi \in \mathcal{L}_0$ and every $t \in T$,*

$$[\![\varphi]\!]_t = K_t \quad \iff \quad [\![\varphi]\!]'_t = K'_t.$$

Proof. Persistence of V': if $s \leq t$ and $\mathrm{val}'(p,s) = K'_s$, then $\mathrm{val}(p,s) = K_s \neq \emptyset$, so persistence of V forces $\mathrm{val}(p,t) = K_t$ and hence $\mathrm{val}'(p,t) = K'_t \supseteq K'_s$. Both valuations are two-valued, so Theorem 4.17 applies on each side and identifies the maps $\varphi \mapsto ([\![\varphi]\!]_t = K_t)$ and $\varphi \mapsto ([\![\varphi]\!]'_t = K'_t)$ with one and the same Kripke forcing relation on $(T, \leq)$, namely the one determined by $t \Vdash^K p \iff \mathrm{val}(p,t) = K_t$. □

Definition 4.19 (The logic of a realising class)**.** Let $\mathcal{C}$ be a realising class in the sense of Definition 5.1. The logic of $\mathcal{C}$ is the set

$$\mathrm{Log}(\mathcal{C}) \;=\; \{\varphi \in \mathcal{L}_0 \;:\; \varphi \text{ is valid on every filtration model whose finite simplicial stage complexes all lie in } \mathcal{C}\},$$

i.e. the set of modality-free propositional formulas valid on every realising class $\mathcal{C}$.

Theorem 4.20 (Geometric Non-Degeneracy is Logically Inert)**.** *Let $\mathcal{G}$ be any condition imposed on the stage complexes $(K_t)_{t\in T}$ alone that is* monotone constructible*: for every filtration $(K_t)_{t\in T}$, there is an effectively obtainable family of complexes $K'_t \supseteq K_t$ such that $K'_s \subseteq K'_t$ whenever $s \leq t$, and every K'_t satisfies $\mathcal{G}$. Examples include $\dim K_t \geq d$, connectedness, contractibility, or strictly increasing vertex sets when these conditions are supplied with such a monotone construction, while the valuation V ranges freely subject only to persistence. Then the set of $\mathcal{L}_0$-formulas valid over all filtration models satisfying $\mathcal{G}$ is exactly IPC.*

Proof. One inclusion is Soundness (Theorem 4.5): every IPC theorem is valid over all filtration models, in particular those satisfying $\mathcal{G}$. For the

converse, let $\varphi \notin$ IPC. By the finite model property choose a finite Kripke countermodel, and by Definition 4.10 its point filtration over $\mathbf{1}$ refutes φ with two-valued atoms. Now invoke the monotone-constructibility hypothesis for $\mathcal{G}$: it supplies an effective family $K'_t \supseteq K_t$ with $K'_s \subseteq K'_t$ whenever $s \leq t$, and each K'_t satisfies $\mathcal{G}$. By Lemma 4.18, inflating the point filtration to these K'_t preserves the induced forcing relation, since the valuation stays two-valued on the distinguished vertex. The inflated model satisfies $\mathcal{G}$ yet still refutes φ; hence no formula outside IPC is $\mathcal{G}$-valid, while Soundness shows no IPC theorem fails. Therefore the $\mathcal{G}$-valid $\mathcal{L}_0$-formulas are exactly IPC. □

Remark 4.21. We claim this only for monotone constructible conditions and only for the language $\mathcal{L}_0$: that is precisely the level at which Lemma 4.18 applies without further hypotheses, and the modal extension of Section 9 shows that the restriction to $\mathcal{L}_0$ is essential.

Remark 4.22 (Logical neutrality of geometric constraints)**.** Theorem 4.20 applies precisely to the $\mathcal{L}_0$-fragment. Within $\mathcal{L}_0$, any monotone-constructible class $\mathcal{G}$ of geometric constraints on the complexes K_t acts as a logical identity: it neither adds nor removes entailments, so the passage from the simplicial semantics of Section 3 to ordinary Kripke models is not a reduction but a conservative embedding, and the geometric layer functions as a verification environment rather than as a constraint on the consequence relation. Over $\mathcal{L}_\Box$ the no-go fails: $\Box\varphi \to \varphi$ is valid on every realising class but not a theorem of IPC, so geometric constraints can then cut down the logic. Theorem 4.20 is therefore best understood as a sharp no-go result delimiting the price of working in a purely propositional language.

5. Realising Classes and Bounded-Depth Logics

The inertness theorem (Theorem 4.20) shows that no geometric hypothesis on the stage complexes can yield a proper superintuitionistic logic. This section turns that negative result into a positive classification by identifying exactly which classes of filtrations realise IPC, and which intermediate logics arise when one constrains the growth of the vertex sets.

5.1. Realising classes

Definition 5.1 (Realising class)**.** Let $\mathcal{C}$ be a class of finite monotone filtrations with nonempty stages. We say that $\mathcal{C}$ is *realising* if for every finite poset $(T, \leq)$ there exist a finite simplicial complex K, the *ambient complex*

of the realisation, and a filtration $(K_t)_{t\in T} \in \mathcal{C}$ with $K_t \subseteq K$ for all $t \in T$, such that the stage map

$$\iota_T \colon (T, \leq) \longrightarrow \big(\mathrm{Sub}(K), \subseteq\big), \qquad \iota_T(t) \;=\; K_t,$$

is an *order-embedding*, i.e. for all $s, t \in T$,

$$s \leq t \quad \Longleftrightarrow \quad K_s \subseteq K_t.$$

Here $\mathrm{Sub}(K)$ denotes the poset of subcomplexes of K ordered by inclusion. Note that an order-embedding is automatically injective, and that $s < t$ implies $K_s \subsetneq K_t$; both K and the filtration $(K_t)_{t\in T}$ are allowed to depend on T.

Remark 5.2 (Ambient complex)**.** The ambient complex K need not be a stage of the filtration. If T has a greatest element $t_\top$ one may always take $K = K_{t_\top}$; in general $K = \bigcup_{t\in T} K_t$ suffices, and this is the canonical choice used throughout. Fixing a single ambient K is what makes the comparison $K_s \subseteq K_t$ a statement inside one fixed subcomplex lattice rather than a comparison of unrelated complexes.

Corollary 5.3 (Frames are realisable)**.** *If $\mathcal{C}$ is realising, then every finite Kripke frame $(T, \leq)$ occurs, up to isomorphism, as the stage structure of a filtration model whose stage complexes belong to $\mathcal{C}$.*

Proof. Immediate from Definition 5.1: the order-embedding ι_T is an isomorphism of $(T, \leq)$ onto its image $\{K_t : t \in T\} \subseteq \big(\mathrm{Sub}(K), \subseteq\big)$, and the accessibility relation of the induced filtration model is exactly inclusion of stages. □

Remark 5.4 (Realising classes are IPC-complete)**.** Let $\mathcal{C}$ be a realising class in the sense of Definition 5.1, and let $\varphi \in \mathcal{L}_0$. The equivalence, φ is valid on every filtration model whose stage complexes form a filtration in $\mathcal{C}$ if and only if $\vdash \varphi$, is established by the No-Go theorem (Theorem 5.8) below, which subsumes this observation.

Remark 5.5 (Non-vacuity and scope)**.** The class of all finite monotone filtrations with nonempty stages is realising: by Lemma 4.6 every finite poset embeds as the stage structure of some monotone filtration. The no-go theorem below is therefore not vacuous, it identifies a genuine limit on what geometry alone can contribute at the propositional level. The only ways past IPC are to leave the class of realising filtrations, as in the bounded-growth constructions of Theorem 5.13, or to enrich the language, as in Section 9.

Example 5.6 (Path versus circle: the reduct forgets H_1)**.** Take $T = \{*\}$ a single stage, so that all filtrations below are constant and Definition 3.5 specialises to a single complex K; write

$$\uparrow\sigma = \{\tau \in K \mid \tau \supseteq \sigma\}, \qquad B(S) = \{\sigma \in K \mid \uparrow\sigma \subseteq S\}$$

for the coface filter and the resulting box-set of Definition 3.5(7). Consider two complexes on $\{a, b, c\}$, the *path*

$$K = \big\{\{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}\big\}, \qquad H_1(K) = 0,$$

and the *circle*

$$K' = \partial\Delta^2 = K \cup \big\{\{a, c\}\big\}, \qquad H_1(K') \cong \mathbb{Z},$$

each carrying the same single atom, valued on the closed star of the edge ab (Figure 1):

$$A := [\![p]\!] = \big\{\{a\}, \{b\}, \{a, b\}\big\}.$$

Both A and its ambient complexes are subcomplexes, so the valuation is legitimate in either model. Below we abbreviate simplices as $a, ab, \ldots$.

The $\Box$-free reduct cannot separate them. Let $\mathcal{L}_0$ denote the $\Box$-free fragment $\{\wedge, \vee, \to, \bot\}$. In *both* models Definition 3.5(5) gives

$$u := [\![\neg p]\!] = \operatorname{int}\big(\complement A\big) = \big\{\{c\}\big\},$$

since $\downarrow bc \ni b \notin \complement A$ and, in K', $\downarrow ac \ni a \notin \complement A$, so both edges are stripped by the interior. Setting $w := A \vee u = \{a, b, c, ab\}$, the subalgebra generated by the single valuation A is

$$\langle A\rangle_{\mathcal{L}_0} = \{\bot,\ u,\ A,\ w,\ \top\}, \qquad \bot < u, A < w < \top,$$

with u, A incomparable; closure is witnessed by $\neg u = A$, $\neg w = \bot$, $A \to u = u$, $u \to A = A$, $w \to A = A$, $w \to u = u$, each computed identically in K and K' because the extra edge ac is removed by int in every case. Hence $\langle A\rangle^{K}_{\mathcal{L}_0} \cong \langle A\rangle^{K'}_{\mathcal{L}_0}$ as Heyting algebras, and *no* $\mathcal{L}_0$-formula distinguishes the path from the circle.

The spatial modality does. The coface test is sensitive to exactly the edge that int discards. In the path,

$$\uparrow a = \{a, ab\} \subseteq A, \qquad \uparrow ab = \{ab\} \subseteq A,$$

while $\uparrow b = \{b, ab, bc\} \not\subseteq A$, so

$$B(A) = \{a, ab\}, \qquad [\![\Box p]\!]_K = \mathrm{int}_K\{a, ab\} = \{a\} \neq \varnothing$$

(the edge ab fails because $\downarrow ab \ni b \notin B(A)$). In the circle the new edge is a coface of a outside A, so a is stripped:

$$\uparrow a = \{a, ab, ac\} \not\subseteq A, \qquad B(A) = \{ab\}, \qquad [\![\Box p]\!]_{K'} = \mathrm{int}_{K'}\{ab\} = \varnothing.$$

Therefore $\neg\Box p$ is valid in K' and refuted in K: the class $H_1(K') \neq H_1(K)$ is invisible to $\mathcal{L}_0$ yet detected by a single modal formula. The general pattern behind this computation, monotonicity, validity of T, and the strictly descending chain $\Box^{n+1}\varphi \leq \Box^n\varphi$, is recorded in Proposition 9.4.



K: path, $H_1 = 0$
$[\![\Box p]\!] = \{a\}$

$K' = \partial\Delta^2$: circle, $H_1 \cong \mathbb{Z}$
$[\![\Box p]\!] = \varnothing$

Figure 1: The atom $[\![p]\!] = \{a, b, ab\}$ (highlighted) in two complexes. The dashed edge ac closes the cycle. The generated $\Box$-free reducts are isomorphic Heyting algebras, but ac is a coface of a lying outside $[\![p]\!]$, which empties $[\![\Box p]\!]$ on the circle.

Remark 5.7 (The reduct forgets structure)**.** Example 5.6 is the local witness of the phenomenon that Theorem 5.8 makes precise. Propositional evaluation collapses a complex into the Heyting subalgebra its valuations generate, and distinct geometric structures can generate *isomorphic* subalgebras; the reduct is therefore not faithful to the face incidence data. The coface quantifier in Definition 3.5(7) reads that discarded data back off the complex. We stress that the isomorphism is claimed only for the subalgebra *generated by the atomic valuations*, not for the full lattice of subcomplexes: the latter recovers the face poset of K and would already separate K from K'.

This result subsumes Theorem 4.20: the local inertness established there is recovered here as the special case of the joint classification, and the additional content of the present theorem lies in the classification it yields.

Theorem 5.8 (No-Go)**.** *For every realising class $\mathcal{C}$, the set of $\mathcal{L}_0$-formulas valid on all filtrations in $\mathcal{C}$ is exactly IPC:* $\mathrm{Log}(\mathcal{C}) = \mathrm{IPC}$.

Proof. Soundness. Every member of $\mathcal{C}$ is a filtration in the sense of Definition 3.1, so $\mathrm{IPC} \subseteq \mathrm{Log}(\mathcal{C})$ by Theorem 4.5.

Completeness. Let φ be an $\mathcal{L}_0$-formula with $\nvdash_{\mathrm{IPC}} \varphi$. By the Kripke completeness of IPC with respect to finite rooted posets [6] there are a finite rooted poset $(W, \leq)$ with root w_0, a monotone valuation $\Vdash^K$, and $w_0 \nVdash^K \varphi$. Since $\mathcal{C}$ is realising, choose $(K_t)_{t \in W} \in \mathcal{C}$ whose stage map is an order-embedding of W. Define a valuation on this filtration by

$$\mathrm{val}(p, t) \;:=\; \begin{cases} K_t, & t \Vdash^K p, \\ \emptyset, & \text{otherwise.} \end{cases}$$

This is persistent (Definition 3.4): if $s \leq t$ and $s \Vdash^K p$ then $t \Vdash^K p$ by monotonicity of $\Vdash^K$, and $K_s \subseteq K_t$ by monotonicity of the filtration. The valuation is two-valued, so Theorem 4.17 gives, for every stage t, every $\sigma \in K_t$ and every $\mathcal{L}_0$-formula ψ,

$$t \Vdash_\sigma \psi \iff t \Vdash^K \psi.$$

Taking $\psi = \varphi$, $t = w_0$ and any $\sigma \in K_{w_0}$ (nonempty by Definition 5.1) yields $w_0 \nVdash_\sigma \varphi$. Hence $\varphi \notin \mathrm{Log}(\mathcal{C})$. □

Scope of the no-go theorem.. Theorem 5.8 concerns the modality-free language $\mathcal{L}_0$ and classes satisfying Definition 5.1, under the hypotheses stated in the theorem. Its conclusion must not be extended to arbitrary classes of geometric models. In particular, a geometric restriction can be covered by this theorem only when the resulting class still satisfies those hypotheses.

Remark 5.9 (The result is fragment-specific)**.** Over $\mathcal{L}_\Box$ the no-go theorem fails: $\Box\varphi \to \varphi$ is valid on every realising class, since $\Box A \subseteq A$ by Proposition 9.4(4), yet it is not a theorem of IPC. Theorem 5.8 applies precisely to the $\mathcal{L}_0$-fragment.

Lemma 5.10 (Padding Lemma)**.** *Let P be an operation on finite simplicial complexes such that*

1. $K \subseteq P(K)$ *(*expansive*);*
2. $K \subseteq K' \Rightarrow P(K) \subseteq P(K')$ *(*monotone*);*

3. $K \subsetneq K' \Rightarrow P(K) \subsetneq P(K')$ *(*strict*).*

If a class $\mathcal{C}$ contains the stagewise image $(P(K_t))_{t\in T}$ of every finite monotone filtration $(K_t)_{t\in T}$ with nonempty stages, then $\mathcal{C}$ is realising, and hence $\mathrm{Log}(\mathcal{C}) = \mathrm{IPC}$.

Proof. Let $(W, \leq)$ be a finite poset. By Lemma 4.6 there is a finite monotone filtration $(K_t)_{t\in W}$ with nonempty stages whose stage map is an order-embedding. By (ii) the family $(P(K_t))_{t\in W}$ is again monotone, by (i) its stages are nonempty, and by (iii) the stage map remains an order-embedding. By hypothesis $(P(K_t))_{t\in W} \in \mathcal{C}$, so $\mathcal{C}$ is realising; the identification of the logic is Theorem 5.8. □

Corollary 5.11 (Geometric properties are inert)**.** *Over $\mathcal{L}_0$, each of the following classes yields exactly IPC.*

1. Connectedness and contractibility. *Take $P(K) = K * \{v\}$, the cone on a fresh vertex v; P is expansive, monotone and strict, and every $P(K)$ is contractible, hence connected.*

2. Lower dimension bounds $\dim \geq m$. *Take $P(K) = K * \Delta^m$ on a fresh vertex set; then $\dim P(K) \geq m$, and P again satisfies Lemma 5.10(i)–(iii).*

3. Any conjunction of the above*, by composing the two joins.*

Note that padding raises dimension, so it cannot be used to realise an upper *dimension bound; classes defined by* $\dim \leq m$ *are treated by the vertex-growth analysis of Theorem 5.13. This answers the question posed in the Introduction: geometry alone cannot move the $\mathcal{L}_0$-logic beyond IPC.*

5.2. Bounding vertex growth: the logics $\mathbf{BD}_n$

Definition 5.12 (Depth and the axioms bd_n)**.** A poset $(T, \leq)$ has *depth* $\leq n$ if it contains no chain $t_1 < t_2 < \cdots < t_{n+1}$. Define

$$\mathrm{bd}_1 \;:=\; p_1 \vee \neg p_1, \qquad \mathrm{bd}_{k+1} \;:=\; p_{k+1} \vee \big(p_{k+1} \to \mathrm{bd}_k\big),$$

and let $\mathbf{BD}_n := \mathrm{IPC} + \mathrm{bd}_n$. Then $\mathbf{BD}_n$ is sound and complete with respect to the Kripke frames of depth $\leq n$, has the finite model property, and is decidable [6].

Theorem 5.13. *Fix $n \geq 1$ and let $\mathcal{C}_n$ be the class of finite monotone filtrations $(K_t)_{t \in T}$ such that*

1. *every stage is nonempty;*
2. *$|V(K_s)| < |V(K_t)|$ whenever $s < t$ (strict vertex growth);*
3. *$|V(K_t)| \leq n$ for all $t \in T$.*

Then

$$\mathrm{Log}(\mathcal{C}_n) \;=\; \mathbf{BD}_n.$$

Proof. Soundness. Let $(K_t)_{t \in T} \in \mathcal{C}_n$. If $t_1 < \cdots < t_k$ is a chain in T, then (ii) makes $|V(K_{t_1})| < \cdots < |V(K_{t_k})|$ a strictly increasing sequence of integers lying in $\{1, \ldots, n\}$ by (i) and (iii); hence $k \leq n$ and $(T, \leq)$ has depth $\leq n$. Fix σ and put $T_\sigma = \{t \in T : \sigma \in K_t\}$, an up-set of T, so $\mathrm{depth}(T_\sigma) \leq n$ as well. The forcing clauses evaluate $t \Vdash_\sigma \cdot$ by quantifying over $\{t' \in T_\sigma : t' \geq t\}$, so $(T_\sigma, \leq)$ together with $t \mapsto [\sigma \in \mathrm{val}(p, t)]$ is an ordinary Kripke model of depth $\leq n$; by Definition 5.12, bd_n is forced there. Since σ and t were arbitrary, $\mathbf{BD}_n \subseteq \mathrm{Log}(\mathcal{C}_n)$.

Completeness. Let $\nvdash_{\mathbf{BD}_n} \varphi$. By the finite model property of $\mathbf{BD}_n$ there are a finite rooted poset $(W, \leq)$ of depth $\leq n$ with root w_0, a monotone valuation $\Vdash^K$, and $w_0 \nVdash^K \varphi$. For $t \in W$ let

$$d(t) \;:=\; \max\{\, k \;:\; \exists\, t_1 < \cdots < t_k = t \text{ in } W \,\},$$

so $1 \leq d(t) \leq n$, and $s < t \Rightarrow d(s) < d(t)$ because any chain ending at s extends by t. Fix vertices $v_1, \ldots, v_n$ and set

$$K_t \;:=\; \mathcal{P}_{\neq \emptyset}\big(\{v_1, \ldots, v_{d(t)}\}\big),$$

the full simplex on the first $d(t)$ vertices together with all its faces. Then $K_t \neq \emptyset$; $s \leq t \Rightarrow K_s \subseteq K_t$, so the filtration is monotone; $s < t \Rightarrow |V(K_s)| = d(s) < d(t) = |V(K_t)|$, giving strict growth; and $|V(K_t)| \leq n$. Hence $(K_t)_{t \in W} \in \mathcal{C}_n$. Equip it with the two-valued valuation $\mathrm{val}(p, t) = K_t$ if $t \Vdash^K p$ and $\emptyset$ otherwise, which is persistent as in the proof of Theorem 5.8. By Theorem 4.17, $t \Vdash_\sigma \psi \iff t \Vdash^K \psi$ for all t, all $\sigma \in K_t$ and all ψ; with $\psi = \varphi$ and $t = w_0$ this refutes φ in a model from $\mathcal{C}_n$. Therefore $\mathrm{Log}(\mathcal{C}_n) \subseteq \mathbf{BD}_n$. $\square$

Relation to the no-go theorem.. The comparison with Theorem 5.8 depends on the hypotheses imposed on the respective model classes. In particular, the no-go conclusion cannot be transferred to the restricted class considered here without checking the realisation requirement and the remaining hypotheses of that theorem.

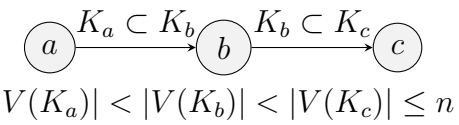


Figure 2: A representative bounded-depth filtration: strict vertex growth forces every chain of stages to have length at most n.

Remark 5.14 (Decidability)**.** Since $\mathbf{BD}_n$ has the finite model property and is decidable [6], Theorem 5.13 transfers decidability to the geometric side: validity of an $\mathcal{L}_0$-formula on $\mathcal{C}_n$ is decidable.

Remark 5.15 (A single classification)**.** Theorems 5.8 and 5.13 are two halves of one statement. Without a constraint on vertex growth the logic is exactly IPC, whatever geometric conditions are imposed on the stages; bounding the vertex sets by n under strict growth pins down exactly $\mathbf{BD}_n$. What moves the logic is combinatorial capacity, not shape.

Remark 5.16 (No-Go vs. $\mathbf{BD}_n$: the dividing line)**.** Theorem 5.8 applies to *realising* classes, i.e. those embedding *every* finite poset. The class $\mathcal{C}_n$ is not realising: a chain $t_1 < \cdots < t_{n+1}$ would require $n+1$ strictly increasing nonempty vertex sets of size $\leq n$, which is impossible. So $\mathcal{C}_n$ escapes Theorem 5.8 exactly by lacking the capacity to realise all posets, and the price of that escape is measured by bd_n.

6. Stage-Local Invariants and Their Limits

6.1. The gap between the two negations

Proposition 6.1. *For any subcomplex $p \in \mathrm{Sub}(K)$:*

1. $\neg_H p \subseteq \neg_{\mathrm{alg}} p$*;*
2. *equality holds iff $\neg_{\mathrm{alg}} p = K \setminus p$ is closed under faces, i.e. iff $\neg_{\mathrm{alg}} p \in \mathrm{Sub}(K)$;*

3. *in general $\neg_H p$ is the largest subcomplex contained in the set-theoretic complement.*

Proof. By Corollary 2.13, $\neg_H p = \bigcup\{r \in \mathrm{Sub}(K) \mid r \cap p = \emptyset\}$. Any such r satisfies $r \subseteq K \setminus p = \neg_{\mathrm{alg}} p$, which gives (i) and (iii). For (ii): if $K \setminus p$ is face-closed it is itself the largest such r, so equality holds; conversely equality forces $K \setminus p = \neg_H p \in \mathrm{Sub}(K)$. □

Example 6.2. Let K be the filled triangle on $\{a, b, c\}$,

$$K = \big\{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\big\},$$

and $p = \{\{a\}, \{b\}, \{a, b\}\}$. Then $\neg_{\mathrm{alg}} p = K \setminus p = \{\{c\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$, which is not a subcomplex: $\{a, c\}$ is present while its face $\{a\}$ is not. Heyting negation keeps only the face-closed part,

$$\neg_H p = \big\{\{c\}\big\},$$

since any subcomplex disjoint from p can contain none of $\{a, c\}, \{b, c\}, \{a, b, c\}$. Hence $\neg_H p \subsetneq \neg_{\mathrm{alg}} p$, the offending simplices being exactly those three.

6.2. Graded negation along a filtration

Definition 6.3 (Graded negation)**.** For a filtration $(K_t)_{t \in T}$ with persistent valuation, and $p_t := \mathrm{val}(p, t)$, define the *graded negation*

$$\neg^{(t)} p \;:=\; \neg_H^{K_t} p_t,$$

the Heyting negation computed inside the single stage K_t.

Proposition 6.4. *If $s \le t$ then $\neg^{(t)} p \cap K_s \subseteq \neg^{(s)} p$.*

Proof. Put $r := \neg^{(t)} p \cap K_s$. Both factors are subcomplexes and $\mathrm{Sub}(K_s)$ is closed under intersection with subcomplexes of K_t, so $r \in \mathrm{Sub}(K_s)$. By persistence (Definition 3.4) $p_s \subseteq p_t$, and $\neg^{(t)} p \cap p_t = \emptyset$ by Corollary 2.13; hence $r \cap p_s \subseteq \neg^{(t)} p \cap p_t = \emptyset$. Thus r is a subcomplex of K_s disjoint from p_s, and maximality of $\neg_H^{K_s} p_s$ gives $r \subseteq \neg^{(s)} p$. □

Remark 6.5 (Reading of the grading)**.** The index in $\neg^{(t)}$ is a filtration stage, not a dimension, and the operator is antitone in it: *truth grows, refutation shrinks*. Being stage-local, $\neg^{(t)}$ need not agree with the interpretation of $\neg$ in the filtration model, which quantifies over all later stages; the discrepancy is exactly the gap isolated in Proposition 6.1, and Subsection 6.4 shows that

the mismatch cannot be repaired. The grading is therefore a diagnostic on the algebras $\mathrm{Sub}(K_t)$, locating the stage at which a refutation is lost; no claim is made that it enlarges the set of validities. The asymmetry is close in spirit to Dummett's account of intuitionistic negation as a claim about all available constructions [11], and the picture of a background shifting while inference proceeds invites comparison with Wittgenstein's river-bed [33, §§96–99]; neither comparison is presupposed below.

6.3. The clarity operator

Definition 6.6 (Clarity)**.** The *clarity operator* on $\mathrm{Sub}(K)$ is

$$\mathrm{Cl}(p) \;:=\; \neg_H \neg_H p.$$

Standard Heyting-algebra facts give $p \subseteq \mathrm{Cl}(p)$, monotonicity, idempotence, and $\mathrm{Cl}(p \cap q) = \mathrm{Cl}(p) \cap \mathrm{Cl}(q)$; hence Cl is not merely a closure operator but a *nucleus*. Its fixed points are the *clear* (regular) subcomplexes.

Proposition 6.7. *The clear subcomplexes form a Boolean algebra, the Booleanisation of* $\mathrm{Sub}(K)$*, with* $\wedge = \cap$*, join* $p \sqcup q := \mathrm{Cl}(p \cup q)$*, and negation* $\neg_H$*.*

Proof. This is the regular-element theorem for Heyting algebras. The SC reading: clarity erases ambiguity at the boundary, since p is clear iff every simplex that cannot be refuted, i.e. lies outside $\neg_H p$, is already present, so nothing hovers between p and its Heyting complement. □

Example 6.8. Continue with K and $p = \{\{a\}, \{b\}, \{a, b\}\}$ from Example 6.2, where $\neg_H p = \{\{c\}\}$. Then

$$\mathrm{Cl}(p) = \neg_H \{\{c\}\} = \bigcup \big\{ r \in \mathrm{Sub}(K) \;\big|\; r \cap \{\{c\}\} = \emptyset \big\} = \big\{\{a\}, \{b\}, \{a, b\}\big\} = p,$$

so p is already clear. A genuine clarification appears for $r = \{\{a\}, \{b\}\}$: again $\neg_H r = \{\{c\}\}$, whence

$$\mathrm{Cl}(r) = \big\{\{a\}, \{b\}, \{a, b\}\big\} \supsetneq r,$$

and the edge $\{a, b\}$ is clarified in: no subcomplex disjoint from r excludes it, so regularisation forces it back.

Remark 6.9. The clarified-in simplices of Example 6.8 occupy a stable third position: neither verified nor refutable, yet not revisable in the manner of the mixed region. Within the algebra they play a role structurally analogous

to the hinge propositions of [33], with the difference that membership is decidable from the face poset. We stress that Cl carries no temporal reading: by Definition 6.6 it is a nucleus computed inside a single stage.

Remark 6.10 (Clarity adds no deductive strength)**.** Cl is definable in $\mathcal{L}_0$, being $\neg\neg$; adding it as a primitive validates no new formulas. Its interest is structural: it exposes the regular subcomplexes and the geometric content of double-negation stability (Proposition 6.7).

6.4. Non-internalisability of stage-local negation

Proposition 6.11. *There is no $\mathcal{L}_0$-formula $\psi(p)$ such that $[\![\psi(p)]\!]_t = \neg^{(t)}p$ in every filtration model at every stage t. More generally, no operator that is strictly antitone in the stage index on some model can be the semantics of a connective.*

Proof. Heredity (Lemma 3.6) requires truth sets to be *monotone* in the stage index: if $s \leq t$ then $[\![\psi]\!]_s \subseteq [\![\psi]\!]_t$. Graded negation violates this. Take $T = \{0 < 1\}$, $K_0 = K_1 = \{\{a\}\}$, and $\mathrm{val}(p, 0) = \emptyset$, $\mathrm{val}(p, 1) = \{\{a\}\}$; this valuation is persistent. Then

$$\neg^{(0)}p = \neg_H^{K_0}\emptyset = K_0 = \{\{a\}\}, \qquad \neg^{(1)}p = \neg_H^{K_1}\{\{a\}\} = \emptyset.$$

If some ψ had $[\![\psi]\!]_t = \neg^{(t)}p$, then $0 \Vdash_{\{a\}} \psi$ while $0 \leq 1$ and $\{a\} \in K_1$, so heredity would give $1 \Vdash_{\{a\}} \psi$, contradicting $[\![\psi]\!]_1 = \emptyset$. Hence no such ψ exists. The general claim follows by the same argument applied to any strictly antitone family. □

Remark 6.12. This is a second, smaller no-go result: no stage-local operator can be a propositional connective of SC-Logic. $\neg^{(t)}$ is best read as a meta-level invariant stratifying a single intuitionistic negation into a persistence-indexed hierarchy. It motivates a genuine modal operator $\Box$ (Section 9), whose heredity is built into its definition rather than imposed pointwise.

7. Robust Topological Queries under Filtration

This section illustrates filtration queries using Heyting negation and the clarity operator $\mathrm{Cl} = \neg_H\neg_H$ of Definition 6.6. The examples contrast the unstable set-theoretic complement with the internal, face-closed operations of SC-Logic.

7.1. The flickering of the classical complement

A set-theoretic complement of a subcomplex need not be face-closed, whereas Heyting negation remains internal to $\mathrm{Sub}(K)$.

Example 7.1 (The set-complement is not a subcomplex)**.** Let K be the filled triangle on vertices $\{a, b, c\}$, i.e.

$$K = \big\{\{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\big\},$$

and let $p = \{\{a\}\}$ be the subcomplex consisting of the single vertex a. The set-theoretic complement is

$$1 \setminus p = \big\{\{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\big\},$$

which is *not* a subcomplex: it contains the edge $\{a, b\}$ but not its face $\{a\}$. By contrast, the Heyting negation (Corollary 2.13) returns the largest subcomplex disjoint from p,

$$\neg_H p = \{\{b\}, \{c\}, \{b, c\}\},$$

which is a genuine subcomplex. Thus $\neg_H p \subsetneq 1 \setminus p$, and only $\neg_H p$ is a legitimate object of the logic.

For a stage t, let $C_t := K_t \setminus \mathrm{val}(p, t)$ and let D_t denote its defect set. The next example records the resulting flicker.

Example 7.2 (Flickering of the set-complement)**.** Let $T = \{1, 2, 3\}$ and let $F = (K_t)_{t \in T}$ be the filtration on the vertex set $\{a, b\}$ given by

$$K_1 = \big\{\{a\}, \{b\}\big\}, \qquad K_2 = K_3 = K_1 \cup \big\{\{a, b\}\big\},$$

with the persistent valuation

$$\mathrm{val}(p, 1) = \mathrm{val}(p, 2) = \big\{\{a\}\big\}, \qquad \mathrm{val}(p, 3) = \big\{\{a\}, \{a, b\}\big\}.$$

Both $t \mapsto K_t$ and $t \mapsto \mathrm{val}(p, t)$ are monotone, and each $\mathrm{val}(p, t)$ is a subcomplex of K_t. The set-complements are

$$C_1 = \big\{\{b\}\big\}, \qquad C_2 = \big\{\{b\}, \{a, b\}\big\}, \qquad C_3 = \big\{\{b\}\big\},$$

whence

$$D_1 = \varnothing, \qquad D_2 = \big\{\{a, b\}\big\}, \qquad D_3 = \varnothing.$$

Indeed C_2 contains the edge $\{a, b\}$ but not its face $\{a\}$, while C_1 and C_3 are genuine subcomplexes. The set-complement therefore *flickers* between well-formed and ill-formed states as the filtration proceeds, so no query built on it can be stable in t.

By Corollary 2.13, Heyting negation always remains in $\mathrm{Sub}(K_t)$, so its defect set is empty.

7.2. A sensor-network filtration

For illustration, take six sensors on a cyclic index set and a three-stage Vietoris–Rips filtration $F = (K_t)_{t\in T}$, $T = \{0, 1, 2\}$, in which a transient one-dimensional hole appears at the middle stage.

$$\{s_{i_0}, \dots, s_{i_k}\} \in K_t \iff \max_{0\le a<b\le k} d_{\mathrm{cyc}}(i_a, i_b) \le t, \qquad d_{\mathrm{cyc}}(i,j) = \min\big(|i-j|, 6-|i-j|\big).$$

- K_0: vertices only.
- K_1: the hexagonal cycle.
- K_2: the cycle filled by the prescribed triangles.

The inclusions $K_0 \subseteq K_1 \subseteq K_2$ are monotone, so this is a filtration in the sense of Definition 3.

$$\{s_{i_0}, \dots, s_{i_k}\} \in K_t \iff \max_{0\le a<b\le k} d_{\mathrm{cyc}}(i_a, i_b) \le t, \qquad d_{\mathrm{cyc}}(i,j) = \min\big(|i-j|, 6-|i-j|\big).$$

Explicitly:

- K_0: the six vertices only (all pairwise distances > 0).
- K_1: the vertices together with the six "short" edges $\{s_i, s_{i+1}\}$, forming the hexagonal cycle ∂H (a 1-dimensional complex with one 1-cycle).
- K_2: all vertices, all edges $\{s_i, s_j\}$ with $d_{\mathrm{cyc}}(i,j) \le 2$, and *all* triangles $\{s_i, s_j, s_k\}$ whose three indices are pairwise within cyclic distance 2. Concretely the triangles are exactly the six "fans" $\{s_{i-1}, s_i, s_{i+1}\}$ together with the two "alternating" triangles $\{s_0, s_2, s_4\}$ and $\{s_1, s_3, s_5\}$.

The inclusions $K_0 \subseteq K_1 \subseteq K_2$ are monotone, so F is a filtration in the sense of Definition 3, and $(T, \le)$ is its filtration frame.

Remark 7.3. The one-dimensional hole of K_1 (the hexagonal loop ∂H) is the topological feature we wish to query. It is *transient*: it is born at $t = 1$ and dies at $t = 2$ once enough triangles fill the disk. A robust query language must report this feature only where it genuinely persists, without spurious appearances caused by non-subcomplex artefacts.

7.3. Robustness: stability and regularity

Recall from Lemma 3.7 that, for a persistent valuation satisfying face-heredity of atoms, every truth set $[\![\varphi]\!]_t$ is a subcomplex of K_t. This is the structural guarantee that Example 7.1 shows a set-theoretic complement cannot provide.

We now formalise two guarantees. The first says that queries built from Heyting negation are *monotone-stable*: once a negated feature is established at a stage, growth of the complex can only shrink its refutation set in a controlled way, never fabricate new cells outside $Sub(K)$. The second says that *clear* queries are exactly the regular (double-negation-stable) subcomplexes, and that clarity is a genuine idempotent that already fixes every Heyting-negated region.

Proposition 7.4 (Monotone stability of the persistent fragment)**.** *Let $p \in Sub(K)$ and let $s \le t$ in T. Then*

$$\neg_H^{K_t} p \;\cap\; K_s \;\subseteq\; \neg_H^{K_s} p.$$

Equivalently, the graded negation $\neg^{(t)}p$ is antitone in t: a cell refuted at stage t that already exists at stage s was refuted at stage s as well. In particular no cell can be refuted, un-refuted, and refuted again along the filtration; the negated fragment does not flicker.

Proof. By Corollary 2.13, $\neg_H^{K_t} p$ is the largest subcomplex of K_t disjoint from p. Let $\sigma \in \neg_H^{K_t} p \cap K_s$. Then $\sigma \in K_s$ and σ lies in a subcomplex of K_t disjoint from p; restricting that subcomplex along the inclusion $K_s \subseteq K_t$ yields a subcomplex of K_s containing σ and still disjoint from p. By maximality of $\neg_H^{K_s} p$ we obtain $\sigma \in \neg_H^{K_s} p$. This is exactly the antitonicity of graded negation, and it forbids the flicker pattern "refuted at s, not at some u with $s \le u \le t$, refuted again at t": the middle failure at u would contradict the inclusion applied to the pair $u \le t$. □

Proposition 7.5 (Regularity of clear queries)**.** *For every $p \in Sub(K)$:*

1. *$C(p) = \neg_H \neg_H p$ is the smallest regular (clear) subcomplex containing p, and $C(C(p)) = C(p)$.*

2. ***(Triple-negation law.)** Every Heyting-negated region is already clear:*

$$\neg_H \neg_H \neg_H p = \neg_H p, \qquad \textit{equivalently} \qquad C(\neg_H p) = \neg_H p.$$

3. *$C(p)$ is disjoint from $\neg_H p$, and $C(p) = \neg_{\mathrm{alg}}\neg_{\mathrm{alg}} p$ holds iff the algebraic double complement is face-closed, i.e. already a subcomplex.*

Proof. (1) By Definition 6.6, $C = \neg_H\neg_H$ is a nucleus on the Heyting algebra $\mathrm{Sub}(K)$: it is inflationary ($p \subseteq C(p)$), monotone, and idempotent. Its fixpoints are the regular elements, and $C(p)$ is the least such element above p.

(2) In any Heyting algebra one has $p \leq \neg_H\neg_H p$; applying the antitone map $\neg_H$ reverses the inequality to $\neg_H\neg_H\neg_H p \leq \neg_H p$. Conversely, instantiating $p \leq \neg_H\neg_H p$ at $\neg_H p$ gives $\neg_H p \leq \neg_H\neg_H(\neg_H p) = \neg_H\neg_H\neg_H p$. The two inequalities force equality $\neg_H\neg_H\neg_H p = \neg_H p$. Reading the left-hand side as $C(\neg_H p)$ yields $C(\neg_H p) = \neg_H p$; thus $\neg_H p$ is a fixpoint of C, i.e. a clear subcomplex, and applying C to it changes nothing.

(3) Since $\neg_H p$ is the largest subcomplex disjoint from p and $C(p) = \neg_H(\neg_H p)$ is disjoint from $\neg_H p$ by the defining adjunction $r \cap (\neg_H p) = \varnothing \iff r \subseteq \neg_H\neg_H p$, disjointness $\mathrm{Cl}(p) \cap \neg_H p = \varnothing$ is immediate. Finally, by Proposition 2.12 the algebraic and Heyting negations coincide exactly. when the algebraic complement is face-closed, giving the stated equality condition. □

7.4. Querying the transient hole

We can now phrase the intended topological query correctly. The naive attempt is to ask for "the covered region and its complementary hole" via $C(\neg c)$, where $c \in Sub(K)$ is the covered subcomplex. By the triple-negation law (Proposition 7.5(2)) this is a trap:

$$C(\neg_H c) = \neg_H\neg_H\neg_H c = \neg_H c,$$

so the clarity pass does nothing and the "interesting" object collapses to the plain negation. The genuinely interesting object is the *stable core of the covered region* itself:

$$\varphi_{\mathrm{core}} := \mathrm{Cl}(c) = \neg_H\neg_H c.$$

The covered region c has this stable core $\mathrm{Cl}(c)$, obtained by stripping boundary-adherent cells that fail to be double-negation stable. Its Heyting complement $\neg_H c$ is the uncovered region; by the triple-negation law $\neg_H\neg_H\neg_H c = \neg_H c$, this region is *itself* regular and requires no further clarity pass.

Concretely, in the filtration of Section 7.2 let $c = K_1$ record "a short communication link exists." Then $\neg_H c$ selects the largest subcomplex carrying *no* short link, and $\mathrm{Cl}(c) = \neg_H\neg_H c$ returns the double-negation-stable

core of the linked region. By Proposition 7.4 the negated fragment $\neg^{(t)}c$ is antitone: as the disk fills at $t = 2$, the uncovered region can only shrink, and it never reappears. The transient hole of K_1 is thus reported precisely on the stages where it is topologically real, with no flickering artefacts. Unlike standard persistence, where stability is a *post hoc* analysis of feature longevity, SC-Logic embeds stability directly into the semantics of the query language, rendering the persistence of the stable core $\mathrm{Cl}(c)$ a primitive logical property.

7.5. Failure of a classical set-complement semantics

Three properties jointly account for the robustness of SC-Logic, none of which is available to a classical set-complement semantics.

1. **Closure inside** $Sub(K)$**.** By Corollary 2.13 and Example 7.1, $\neg_H$ always returns a subcomplex, whereas $1 \setminus p$ generally does not. Every intermediate query is therefore a legitimate topological object.

2. **Monotone stability.** By Proposition 7.4, refutations are antitone along the filtration, so no feature flickers.

3. **Idempotent clarity.** By Proposition 7.5, $C = \neg_H \neg_H$ is an idempotent nucleus whose fixpoints are exactly the regular subcomplexes, and every negated region is already a fixpoint. Clarity acts as a *logical persistence filter*.

Because SC-Logic is decidable and enjoys the finite model property (Proposition 4.15 and Corollary 4.16), each of these queries is effectively computable on the finite complexes K_t. SC-Logic thus provides exactly the guarantees a robust topological query language requires: validity, stability, and computability, in a single Heyting-algebraic package.

8. Comparison with Related Frameworks

The semantics developed above sits at the intersection of three traditions: topological semantics for intuitionistic logic, qualitative spatial reasoning, and the use of simplicial complexes in logic and in topological data analysis. We record precisely how the present framework relates to each, and in particular how Theorem 4.20 should be read against them.

Topological semantics for IPC.. Interpreting intuitionistic connectives by open sets of a topological space goes back to Tarski [30] and to McKinsey and Tarski [23], who proved completeness of IPC with respect to the open-set algebra of any separable metric space that is dense-in-itself, in particular of $\mathbb{R}^n$. Our Heyting algebra $\mathrm{Sub}(K)$ is *combinatorial* rather than point-set: its elements are subcomplexes, a family closed under arbitrary unions and intersections but not equal to the opens of a topology on the underlying vertex set, and the induced negation $\neg_H$ is computed by coface removal rather than by taking the interior of a set-complement (Proposition 2.12, Corollary 2.13). The practical gain is effectivity: implication, negation and clarity are finite face-poset computations, whereas the classical topological models are infinite. The price, made precise by Theorem 4.20, is that this extra structure buys no extra validities.

Kripke frames and Esakia duality.. Kripke [20] gave the relational semantics for IPC in which formulas denote upward-closed sets of a poset, and Esakia duality [12] identifies Heyting algebras with certain ordered topological spaces. Filtration frames (Definition 3.3) are Kripke frames of a restricted shape, stages organised by a monotone filtration parameter, and persistent valuations (Definition 3.4) are exactly the valuations for which truth is hereditary along the filtration (Lemma 3.6). Two points are worth isolating. Restricting to this shape does not restrict the logic: the point-filtration construction of Definition 4.10 converts an arbitrary countermodel into a filtration countermodel, which is what makes strong completeness (Theorem 4.8) go through. Conversely, the face-poset representation (Theorem 2.15) is a *realisability* statement in the opposite direction: not every finite Heyting algebra arises as some $\mathrm{Sub}(K)$, so the class of subcomplex algebras is properly smaller than the class of finite Heyting algebras even though it generates the same logic (Corollary 2.17).

Qualitative spatial reasoning.. Region-based spatial calculi such as RCC [26] and the modal spatial logics surveyed by Aiello, van Benthem and Bezhanishvili [1] also reason about regions rather than points, and also treat complementation with care. The difference is one of aim. Those systems fix a space and axiomatise a repertoire of qualitative relations (connection, tangency, parthood) whose logic is typically modal or first-order. Here the space is not fixed: the object of study is a filtration, the intended queries are monotone in the filtration parameter, and the qualitative content is carried by graded

negation (Definition 6.3) and by regularity (Proposition 7.5) rather than by primitive spatial relations. In particular the flickering phenomenon of Section 7 has no counterpart in the fixed-space setting, since it is precisely an artefact of complementation interacting with growth.

Clarity, nuclei and lax logic.. The clarity operator Cl (Definition 6.6) is defined as $\neg_H \neg_H$ and satisfies more than the axioms of a closure operator: it is a *nucleus* on the Heyting algebra $\mathrm{Sub}(K)$, i.e. an idempotent inflationary interior operator that commutes with meets. Nuclei on Heyting algebras correspond to quotient Heyting algebras via their fixed-point sublattice, and provide the algebraic counterpart of open geometric morphisms in topos theory [19]. In the propositional setting, nuclei interpret the Lawvere-style "lax" modalities studied by Fairtlough and Mendler [13] in the context of resource-aware reasoning, where the modality stratifies formulas into levels of commitment or refinement. Our use of clarity is formally analogous: clear subcomplexes are the stable geometric cores under iterated complementation, and the nucleus structure ensures that the sublattice of clear elements inherits the Heyting structure. This connection suggests that the graded negations of Section 6.2 admit a lax-logic reading, though we do not pursue that interpretation here.

Intuitionistic modal logics.. The operator $\Box$ introduced in Section 9 is a normal, reflexive, non-transitive modality, i.e. an intuitionistic T-modality in the sense of the systems surveyed below. Božić and Došen [4] and Fischer Servi [14] introduced the standard bi-relational Kripke semantics for normal intuitionistic modal logics; Simpson's thesis [29] develops the proof-theoretic and semantic foundations, Ono [24] surveyed the landscape of intuitionistic modal extensions, and Wolter and Zakharyaschev [34] give the modern algebraic and correspondence-theoretic account. Our $\Box$ validates necessitation, axiom K and axiom T (Proposition 9.4) but fails axiom $\mathsf{4}$ (Proposition 9.10), so it is strictly weaker than the intuitionistic $\mathsf{iS4}$ systems of Božić and Došen [4]. We do not claim that SC-Logic's $\Box$ coincides with any of these systems; the comparison is limited to the shared structural properties (normality, reflexivity, non-transitivity), and the completeness question for $\mathcal{L}_\Box$ remains open (Section 9).

Simplicial complexes in epistemic logic.. Simplicial complexes have recently been used as models of distributed knowledge, where vertices are local states of processes and facets are global configurations [16, 10]. Despite the shared

combinatorial substrate, the semantics is orthogonal to ours in two respects. Those models are *modal and classical*: formulas denote sets of facets, the connectives are Boolean, and the logical content lies in the knowledge operators induced by the chromatic structure. Our models are *intuitionistic and non-modal*: formulas denote subcomplexes, the connectives are the Heyting operations of $\mathrm{Sub}(K)$, and no epistemic reading is intended. The two frameworks are therefore best seen as complementary uses of the same geometry, and combining them, knowledge operators over a filtration of chromatic complexes, is a natural question that we do not address here.

Filtrations and topological data analysis.. The filtrations of Definition 3.1 are the standard input of persistent homology [7], and the presheaf-like behaviour noted in Remark 3.2 is the same monotonicity that makes persistence modules well behaved [5]. The invariants, however, are of different type. Persistent homology extracts *homological* invariants, barcodes of cycles, and is insensitive to the lattice structure of subcomplexes. Our graded negations extract *order-theoretic* invariants of the same filtration: the stage from which a query region is persistently avoided, and the stable core of a query under the clarity nucleus (Proposition 7.4). Nothing here computes or approximates a barcode, and nothing in persistence computes $\neg_H$; the two are independent readings of one filtration, and their interaction is left open.

What the inertness theorem does and does not say.. Because the comparisons above may suggest that a richer geometry should yield a richer logic, we state the delimitation explicitly. Theorem 4.20 says that the set of formulas valid on *all* filtration models is exactly IPC, and Theorem 4.17 shows that the standard non-degeneracy conditions on complexes do not change this. It does *not* say that the geometry is irrelevant: it says that the geometry's contribution is invisible to the propositional consequence relation and must be accessed by one of three routes: additional *language*, graded or persistence-indexed modalities, constraints on valuations, or restriction of the class of realising complexes to those satisfying geometric growth conditions, the last of which yields the intermediate logics $\mathbf{BD}_n$ (Theorem 5.13, Corollary 5.11). All three routes are discussed as open problems in Section 10.

9. The Spatial Modality $\Box$

Remark 9.1 (Scope of the modal section)**.** This section is exploratory: soundness and the stated validity results are proved, but a completeness

theorem for the full modal language is not proved here.

The non-internalizability of stage-local negation (Proposition 6.11) motivates a genuine modal operator $\Box$ whose heredity is built into its definition. This section introduces $\Box$ over the face poset, records its basic properties, and states the main technical constraint imposed by padding.

9.1. Definition and heredity

Definition 9.2 (Spatial modality $\Box$)**.** Let $(K_t)_{t\in T}$ be a monotone filtration. For $\sigma \in K_{t'}$ write

$$\uparrow_{t'}\sigma = \{\tau \in K_{t'} \mid \tau \supseteq \sigma\}, \qquad \downarrow\sigma = \{\tau \neq \varnothing \mid \tau \subseteq \sigma\}$$

for the coface filter of σ in $K_{t'}$ and its face closure; note that $\sigma \in \uparrow_{t'}\sigma$. For a formula φ, define the *box-set* at stage t by

$$B_t(\varphi) := \{\sigma \in K_t \mid \forall t' \geq t : \uparrow_{t'}\sigma \subseteq [\![\varphi]\!]_{t'}\},$$

and set

$$[\![\Box\varphi]\!]_t := \mathrm{int}_{K_t}\big(B_t(\varphi)\big), \qquad \mathrm{int}_K(S) := \{\sigma \in K \mid \downarrow\sigma \subseteq S\}.$$

Thus $\sigma \in B_t(\varphi)$ iff φ holds at σ itself (take $t' = t$, $\tau = \sigma$) and at every coface of σ occurring at any future stage. The operator int_{K_t} is the interior of the Alexandrov (down-set) topology on the face poset $\mathcal{F}(K_t)$, so $[\![\Box\varphi]\!]_t$ is the largest subcomplex of K_t contained in $B_t(\varphi)$.

Lemma 9.3 (Persistence of $\Box$)**.** *Let $s \leq t$ and $\sigma \in [\![\Box\varphi]\!]_s$. Then $\sigma \in [\![\Box\varphi]\!]_t$.*

Proof. Assume $\sigma \in [\![\Box\varphi]\!]_s = \mathrm{int}_{K_s}(B_s(\varphi))$. There exists a subcomplex $U \subseteq K_s$ with $\sigma \in U \subseteq B_s(\varphi)$. Since $s \leq t$, the set of future stages shrinks: $\{t' \geq t\} \subseteq \{t' \geq s\}$, so $B_s(\varphi) \subseteq B_t(\varphi)$. Hence $U \subseteq K_s \subseteq K_t$ is a subcomplex of K_t containing σ and contained in $B_t(\varphi)$, giving $\sigma \in \mathrm{int}_{K_t}(B_t(\varphi)) = [\![\Box\varphi]\!]_t$. $\square$

9.2. Basic properties

Proposition 9.4 (Modal profile of $\Box$)**.** *Let $(K_t)_{t\in T}$ be a monotone filtration and let $\varphi, \psi \in \mathcal{L}_\Box$. Then $\Box$ is a* normal *modality validating* T *but not* $\mathsf{4}$*; semantically it is a monotone, deflationary and* non-idempotent *operator on* $\mathrm{Sub}(K_t)$*, hence not an interior operator. Precisely:*

1. (Necessitation) *If φ is valid on the class of filtration models, then so is $\Box\varphi$. Indeed validity of φ gives $B_t(\varphi) = K_t$ for every t, hence $[\![\Box\varphi]\!]_t = \mathrm{int}_{K_t}(K_t) = K_t$. In particular $\Box\top$ is valid.*

2. (Distribution over $\wedge$) $\Box(\varphi \wedge \psi) \leftrightarrow \Box\varphi \wedge \Box\psi$ *is valid: $B_t(\varphi \wedge \psi) = B_t(\varphi) \cap B_t(\psi)$, and int_{K_t}, being the largest-down-set operator on the face poset $\mathcal{F}(K_t)$, preserves finite intersections.*

3. (Axiom K) $\Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)$ *is* valid.

4. (Axiom T) $\Box\varphi \to \varphi$ *is valid: $B_t(\varphi) \subseteq [\![\varphi]\!]_t$ by taking $t' = t$ and $\tau = \sigma$, and int_{K_t} is deflationary.*

5. (Failure of $\mathsf{4}$) $\Box\varphi \to \Box\Box\varphi$ *is* not *valid; this is proved separately in Proposition 9.10. Hence the $\Box$-fragment is not an $\mathsf{S4}$-modality. The failure is* not *caused by the accessibility relation, which is transitive: if $\sigma \in B_t(\varphi)$ and $\tau \supseteq \sigma$ in $K_{t'}$ for some $t' \geq t$, then every coface of τ in $K_{t''}$ with $t'' \geq t'$ is a coface of σ with $t'' \geq t$, so $\tau \in B_{t'}(\varphi)$; that is, the raw coface operator satisfies $B_t(\varphi) \subseteq B_t\big(B(\varphi)\big)$ and would validate $\mathsf{4}$ on its own. Transitivity is destroyed only by the interposed int_{K_t}, which shrinks $B_t(\varphi)$ before the second coface test is applied. Axiom $\mathsf{4}$ is therefore traded for the closure under faces that makes $\Box$ intuitionistically admissible. A concrete instance of this strictly descending chain is computed in Example 5.6, where the extra edge of the circle is detected precisely at the second stage $\Box^2$.*

6. (Deflationary iteration; non-idempotence) $\Box\Box\varphi \to \Box\varphi$ is *valid: it is the instance of item (4) at the formula $\Box\varphi \in \mathcal{L}_\Box$, so $[\![\Box\Box\varphi]\!]_t \subseteq [\![\Box\varphi]\!]_t$ at every stage. Combined with item (5) the inclusion is in general* strict*, whence $\Box\Box \not\equiv \Box$. Thus iterating $\Box$ produces a descending chain*

$$[\![\varphi]\!]_t \;\supseteq\; [\![\Box\varphi]\!]_t \;\supseteq\; [\![\Box\Box\varphi]\!]_t \;\supseteq\; \cdots$$

   of subcomplexes of K_t, each iteration discarding the simplices whose stability under coface extension has just been refuted. The modal depth of a formula is consequently semantically significant: $\Box^n$ is a genuinely finer predicate than $\Box$ for each $n \geq 2$, in contrast with the topological reading of $\mathsf{S4}$ where all iterations collapse.

7. (Monotonicity) *If $\varphi \to \psi$ is valid then so is $\Box\varphi \to \Box\psi$, since B_t and int_{K_t} are both monotone.*

Proof. Items (1), (2), (4), (5) and (6) are proved in the statement. We prove (3).

Fix $t \in T$ and put $X = B_t(\varphi \to \psi)$, $Y = B_t(\varphi)$, $Z = B_t(\psi)$. We claim $X \cap Y \subseteq Z$. Let $\sigma \in X \cap Y$, let $t' \geq t$ and let $\tau \supseteq \sigma$ be a simplex of $K_{t'}$. By $\sigma \in X$ we have $\tau \in [\![\varphi \to \psi]\!]_{t'}$, and by $\sigma \in Y$ we have $\tau \in [\![\varphi]\!]_{t'}$. Since $\mathrm{Sub}(K_{t'})$ is a Heyting algebra (Proposition 2.11), residuation gives

$$[\![\varphi]\!]_{t'} \cap \big([\![\varphi]\!]_{t'} \Rightarrow [\![\psi]\!]_{t'}\big) \;\subseteq\; [\![\psi]\!]_{t'},$$

whence $\tau \in [\![\psi]\!]_{t'}$. As t' and τ were arbitrary, $\sigma \in Z$, proving the claim.

Applying int_{K_t} and using preservation of finite intersections together with monotonicity,

$$[\![\Box(\varphi \to \psi)]\!]_t \cap [\![\Box\varphi]\!]_t = \mathrm{int}_{K_t}(X) \cap \mathrm{int}_{K_t}(Y) = \mathrm{int}_{K_t}(X \cap Y) \subseteq \mathrm{int}_{K_t}(Z) = [\![\Box\psi]\!]_t.$$

The stage t was arbitrary, so the inclusion holds at every stage. Since $\mathrm{Sub}(K_t)$ is a Heyting algebra whose meet is intersection, residuation converts it into

$$[\![\Box(\varphi \to \psi)]\!]_t \;\subseteq\; [\![\Box\varphi]\!]_t \Rightarrow [\![\Box\psi]\!]_t \;=\; [\![\Box\varphi \to \Box\psi]\!]_t,$$

whence $[\![\Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)]\!]_t = K_t$. As t and the model were arbitrary, K is valid. □

Remark 9.5 (The $\mathsf{S4}$-core of $\Box$)**.** Although $\Box$ is not idempotent, the descending chain $\varphi \geq \Box\varphi \geq \Box^2\varphi \geq \cdots$ is monotone and, over a finite complex K_t, stabilises after finitely many steps. Writing

$$\Box^\infty\varphi \;:=\; \bigwedge_{n \geq 0} \Box^n\varphi \;=\; \Box^N\varphi \quad \text{for } N \gg 0,$$

the operator $\Box^\infty$ is idempotent, deflationary and finite-meet preserving, hence an interior operator in the strict sense. It therefore carries a genuine $\mathsf{S4}$-modality on the same Heyting algebra. The point of Proposition 9.4 is that $\Box^\infty$ is strictly coarser than $\Box$: the intermediate stages $\Box^n$, discarded by any $\mathsf{S4}$ treatment, are exactly what separates the two complexes of Example 5.6.

Remark 9.6 (Decidability scope)**.** The validity problem for $\mathcal{L}_0$-formulas on finite filtration models is decidable: the finite model property (Proposition 4.15) bounds the countermodel size by $2^{|\mathrm{Sub}(\varphi)|}$, and model checking on a finite filtration model is effective. By contrast, axiomatic completeness and validity for the full $\mathcal{L}_\Box$-fragment are *open*: no $\Box$-completeness theorem is available (Section 9), and we do not know whether $\mathcal{L}_\Box$-validity is decidable.

Remark 9.7. The apparent counterexample to K obtained from a path a–b–c with $[\![\varphi]\!] = \{a, ab, b\}$ and $[\![\psi]\!] = \{b, bc, c\}$ compares $\Box(\varphi \to \psi)$ with $\Box\varphi \wedge \Box\psi$, which is not the axiom K; it shows only that $\Box$ does not distribute over implication as an *equivalence*. The essential point in Proposition 9.4(3) is that $[\![\varphi \to \psi]\!]$ is the *Heyting* implication of $\mathrm{Sub}(K_{t'})$, not the set-theoretic one, so residuation is available stage-wise.

9.3. Non-definability in $\mathrm{Sub}(K)$

Lemma 9.8 (Non-definability in $\{\wedge, \vee, \to, \bot\}$)**.** *The operator $\Box$ is not definable from the Heyting operations alone.*

Proof. Consider the path $a{-}b{-}c$ (edges ab, bc) as a single-stage complex K. Let $[p] = \{a, ab, b\}$. Then $B(p) = \{\sigma \mid \text{all cofaces of } \sigma \text{ are in } [p]\}$. The only coface of a is ab, which lies in $[p]$; the coface bc of b does *not* lie in $[p]$, so $b \notin B(p)$. The simplex ab has no proper coface, so $ab \in B(p)$. Thus $B(p) = \{a, ab\}$, which is not face-closed since $b \notin \{a, ab\}$, so $[\![\Box p]\!] = \mathrm{int}_K(\{a, ab\}) = \{a\}$. The subalgebra generated by $\{p\}$ under $\{\wedge, \vee, \to, \bot\}$ is $\{\varnothing, \{c\}, p, p \vee \{c\}, K\}$, which does not contain $\{a\}$. Therefore $\Box$ is not definable from the Heyting operations. $\square$

Remark 9.9. Theorem 4.20 and Lemma 9.8 together isolate two distinct roles played by the simplicial structure. In $\mathcal{L}_0$ the geometry yields no validity beyond IPC; nevertheless it supports a well-defined operator that the Heyting operations cannot express. The contrast is reminiscent of the distinction between what a notation states and what it exhibits [32, 4.1212]. We record the comparison as an interpretive observation; nothing below depends on it.

9.4. The axiom T holds; axiom 4 fails

Proposition 9.10. *The operator $\Box$ validates the modal axiom T ($\Box\varphi \to \varphi$) but* not *the axiom $\mathsf{4}$ ($\Box\varphi \to \Box\Box\varphi$).*

Proof. Validity of T is Proposition 9.4(4).

For the failure of $\mathsf{4}$ we exhibit a single-stage filtration. Let K be the path with vertices a, b, c and edges ab, bc, and let the filtration be constant, $K_t = K$ for all $t \in T$. With a single stage, Definition 9.2 reduces to

$$B(A) = \{\sigma \in K \mid {\uparrow}\sigma \subseteq A\}, \qquad [\![\Box\varphi]\!] = \mathrm{int}_K\big(B([\![\varphi]\!])\big),$$

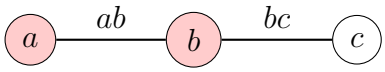
where ${\uparrow}\sigma = \{\tau \in K \mid \tau \supseteq \sigma\}$ is the set of cofaces of σ (including σ itself) and $\mathrm{int}_K(S) = \{\sigma \mid {\downarrow}\sigma \subseteq S\}$ is the largest subcomplex of K contained in S.

Let $[\![p]\!] = \{a, b, ab\}$. This set is face-closed, hence a legitimate element of $\mathrm{Sub}(K)$.

Step 1: $[\![\Box p]\!] = \{a\}$. We compute $B([\![p]\!])$ simplex by simplex: $\uparrow a = \{a, ab\} \subseteq [\![p]\!]$; $\uparrow b = \{b, ab, bc\} \not\subseteq [\![p]\!]$ since $bc \notin [\![p]\!]$; $\uparrow c = \{c, bc\} \not\subseteq [\![p]\!]$; $\uparrow ab = \{ab\} \subseteq [\![p]\!]$; $\uparrow bc = \{bc\} \not\subseteq [\![p]\!]$. Hence $B([\![p]\!]) = \{a, ab\}$, which is not face-closed because $b \notin \{a, ab\}$. Therefore $[\![\Box p]\!] = \mathrm{int}_K(\{a, ab\}) = \{a\}$.

Step 2: $[\![\Box\Box p]\!] = \varnothing$. Here $[\![\Box\Box p]\!] = \mathrm{int}_K\big(B(\{a\})\big)$. Since $\uparrow a = \{a, ab\} \not\subseteq \{a\}$ and every other simplex of K has a coface outside $\{a\}$, we get $B(\{a\}) = \varnothing$, so $[\![\Box\Box p]\!] = \varnothing$.

Step 3. Validity of $\Box p \to \Box\Box p$ requires $[\![\Box p]\!] \subseteq [\![\Box\Box p]\!]$, i.e. $\{a\} \subseteq \varnothing$, which is false. Hence $[\![\Box p \to \Box\Box p]\!] \neq K$ and $\mathsf{4}$ fails. □



$[\![p]\!] = \{a, b, ab\}$
$B([\![p]\!]) = \{a, ab\}, \quad \Box p = \{a\}$
$B(\{a\}) = \varnothing, \quad \Box\Box p = \varnothing$

Figure 3: The three-vertex path countermodel witnessing failure of axiom $\mathsf{4}$.

Remark 9.11. The failure of $\mathsf{4}$ is structural rather than accidental: $[\![\Box\varphi]\!]_t$ is obtained by composing the *upward* closure test B_t on the face poset with the *downward* operator int_{K_t}. Neither factor is idempotent on the range of the other, so $\Box$ is not a topological interior operator and the $\Box$-fragment is a normal T-logic strictly weaker than $\mathsf{S4}$.

Remark 9.12 (Failure of 4 is spatial, not temporal)**.** The countermodel of Proposition 9.10 uses a single fixed stage (a constant filtration), so non-transitivity is not an artifact of the temporal or filtration component. It arises from composing two opposite-direction operators on the face poset: B_t is an upward test over cofaces, and int_{K_t} is a downward closure-interior operator. Hence $\Box$ is not an idempotent topological interior, and the failure of $\mathsf{4}$ is a property of the spatial geometry alone.

9.5. Padding constraint

Remark 9.13 (Padding constraint for $\Box$)**.** Since $B_t(\bot) = \emptyset$ and $B_t(\top) = K_t$, the two-valued subalgebra $\{\emptyset, K_t\}$ is $\Box$-closed: $\Box\emptyset = \emptyset$ and $\Box K_t = K_t$. Hence on extremal (two-valued) valuations, $\Box\varphi \leftrightarrow \varphi$ is valid, and padded

models cannot refute $\Box$-formulas. A canonical model for $\Box$-completeness must therefore use *non-extremal* valuations. This is the main technical constraint: padding does not break $\Box$-validity, it becomes *insufficient* to refute $\Box$-formulas. We leave $\Box$-completeness w.r.t. abstract bi-relational simplicial frames and a separate $\Box$-representation theorem as explicitly open problems.

Remark 9.14 ($\Box$-completeness as open problem)**.** Completeness for the $L_{\Box}$-fragment relative to abstract bi-relational simplicial frames is left as an open problem for Paper III.

10. Conclusion

This paper has grounded SC-Logic in a rigorous semantic foundation, closing the gap that Paper I left open between geometric intuition and formal justification. Four results constitute the core. First, the lattice of subcomplexes of any finite simplicial complex carries a canonical Heyting algebra structure, with the implication given explicitly as the largest subcomplex whose meet with the antecedent is contained in the consequent (Proposition 2.11, Corollary 2.13). Second, monotone filtrations induce Kripke frames with persistent valuations, and the simplex-level forcing semantics agrees with the algebraic one over discrete filtrations: truth sets compute exactly the Heyting operations (Proposition 3.8). Third, the Hilbert-style proof system is sound and complete with respect to the class of filtration models (Theorems 4.5 and 4.8) and enjoys the finite model property (Proposition 4.15), from which decidability follows immediately (Corollary 4.16). Fourth, moving beyond pure IPC, we isolated the gap between algebraic and Heyting negations (Proposition 6.1), introduced graded negation along a filtration with its antitone behaviour (Proposition 6.4), and defined the clarity operator as the double-negation nucleus, whose fixed points form the Booleanization of the subcomplex algebra (Proposition 6.7).

The relation between this paper and Paper I is one of foundation to construction. Paper I established truth-as-geometry operationally: propositions as subcomplexes, inference as topological transformation. The present paper supplies the missing semantics that explain *why* those operations behave intuitionistically. The Heyting algebra of subcomplexes is not a choice of interpretation; it is the canonical structure that the geometry forces. Graded negation makes this precise at the stage level: each filtration step carves out its refutable region, and the erosion of that region as the complex grows is captured algebraically rather than left as geometric intuition.

Taken together, the results identify the propositional, modality-free fragment of SC-Logic exactly with IPC. This identification is not a limitation but a calibration: it tells us precisely what the simplicial-complex geometry is computing when connectives are interpreted topologically, and it marks the boundary at which intuitionistic logic ends and genuinely geometric content begins.

Further directions.. Several directions remain open for Paper III. The most pressing is $\Box$-completeness: soundness for the $\mathcal{L}_{\Box}$-fragment relative to abstract bi-relational simplicial frames is established, but a representation theorem and full completeness proof remain open (Section 9); this is explicitly positioned as an open problem rather than a relied-upon result. A second direction concerns robustness quantification: measuring the persistence interval over which a formula is forced provides a natural complexity measure for inference, and bottleneck-distance machinery from persistent homology can be brought to bear on logical consequence directly. Third, relaxing the monotonicity assumption on filtrations to allow zigzag filtrations would model revision and retraction of geometric evidence, requiring a forcing semantics over zigzag persistence modules. Finally, the computational theory deserves systematic treatment: the complexity of deciding forcing on finite filtrations, and a computational implementation of this framework, sketched in Paper I.

The broader program is a geometric reasoning architecture: logical systems whose connectives are geometric operations, whose models are evolving spaces, and whose inferences carry measurable robustness. The results here show that the propositional, modality-free fragment of spatial reasoning over finite simplicial complexes is exactly intuitionistic logic, no more, no less, and that everything beyond it is geometry.

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